The Ultimate Casebook

FOR PRODUCT MANAGEMENT INTERVIEWS

SYSSOC

SYSTEMS & OPERATIONS SOCIETY, FMS DELHI

How would you design a VR product for hospitals?

Question

Suppose Oculus is thinking of making a VR product for hospitals. How would you design this product?

How to structure your answer?

A good strategy to tackle this sort of question is to use the CIRCLES Method™ from Lewis Lin's book *Decode and Conquer*. The method can be summarized as:

C — Ask **clarifying** questions to narrow the scope (Context).

I — **Identify** the users/customers as personas like food lovers, soccer moms, etc.

R — **Report** on their needs. Describe their behaviors and based on those behaviors what are their needs and describe use cases.

C — **Cut** through and prioritize needs you will address.

L — **List** at least three solutions.

E — **Evaluate** these solutions' tradeoffs

S — **Summarize**:

- Which solution you would recommend
- Recap what the solution does and why is it beneficial
- Why you prefer this solution

Answer Example

INTERVIEWEE: I don't know much about VR technology, except that it is a visual technology that immerses the user into a virtual world, am I right?

INTERVIEWER: Yes, that is correct. There are currently several uses for VR technology, such as diagnosis of brain related injuries such as concussions, Alzheimer's among others, psychiatric treatment, and it is used for entertainment. We are interested in developing a VR product for the case of psychiatric treatment, such as for Post-traumatic stress disorder (PTSD). With VR a person can be taken back to a traumatic event to resurface trauma that can be treated.

INTERVIEWEE: Okay, so, my understanding is that you would like to design a VR product that a psychiatrist could use in a hospital setting, and that this product will help resurface trauma that will be triggered via VR scenes.

INTERVIEWER: Yes.

INTERVIEWEE: Are there any specific requirements or constraints?

INTERVIEWER: Well, it has to be something the patients can put on and feel comfortable wearing while seated or lying down. And of course, cost is an issue.

INTERVIEWEE: Okay, please give me a minute to think about this.

(After one minute.)

INTERVIEWEE: Okay, I would like to start first by discussing who the possible users of this product will be, then possible use case scenarios, a few solutions and wrap up with my recommendation.

Use Cases

INTERVIEWEE: Let's talk about the users. Victims of trauma can be of any age, children or adults. They can be victims of physical or verbal abuse, or have experienced trauma in their line of work like in the case of military or police personnel. So, I see several use cases that arise:

- A child that experienced a traumatic event but cannot recollect it
- Military person that suffered war trauma
- Police that suffered trauma during a violent event
- A civilian that experienced a horrific scene

INTERVIEWEE: In all of these cases, I can see the need for different types of tasks that the product should provide, such as:

- Display scenes similar to the traumatic event. Selection of the scenes should be controlled by the doctor.
- Ability to plan the sequence of different scenes.
- Playback should be easily controlled by the patient and doctor.
- The ability for the doctor to mark and associate comments to specific scenes.
- The ability to transcribe these comments to text for the doctor to study later.
- The ability to search for these marks/comments.
- Maybe the ability to map emotional reactions to different areas of the brain for later analysis.
- The VR headset should be adjustable to children and adult head sizes, or have two different sizes

Prioritization

INTERVIEWEE: Of all the users, I would focus on military personnel because government funding is likely to be available, and there is already an existing large number of war veterans in need.

Solutions

INTERVIEWEE: I think most of the cases I mentioned can be easily added to a VR headset, except for the mapping of emotions to brain areas, since that would require sophisticated sensors. I think that the most pressing need for the psychiatrist is to resurface the trauma or traumatic event they want to treat as opposed to researching deeper what parts of the brain are affected. Therefore, I would recommend focusing on all the cases I mentioned except for the brain mapping feature.

INTERVIEWEE: Here is a list of solutions:

- A pair of VR headsets, one for the patient and one for the doctor that are synchronized when they play, so that the doctor can relate to what the patient is experiencing to understand his condition better.
- The VR headset should be wireless enabled and light so it is easy to wear and move around.
- A handheld wireless remote control for the patient and doctor, so that the doctor can control playback, and the patient could also control it.
- Software for the doctor to create video sequences to present to the patient. A doctor could select videos made by experts that have been curated and associated with typical traumas.
- The ability to load or stream the VR scenes to the headset.
- A digital pad, that is overlaid on the video so the doctor can write annotations on the scenes.
- A transcription of the annotations to text for the doctor.

INTERVIEWEE: For an MVP, I would prioritize the ability to create the video sequences, the remote control feature, and the use of streaming instead of loading into the VR headset. I would leave the annotation and pad for later, because it seems to me that we need to test first if the doctors find this way of treatment useful before adding more functionality. Adding more functionality would only add value if the doctors find the VR display of scenes useful.

INTERVIEWEE: The risk of delaying the annotation feature is that if doctors found the device really useful, they may feel frustrated by not having the ability to make annotations.

INTERVIEWEE: So, to wrap up:

- I recommend making two VR headsets: one for the patient and one for the doctor that are synchronized. And add the remote control functions, video streaming, and software for content creation.
- This solution would allow a doctor to resurface trauma that otherwise would be difficult to do since patients repress traumatic events unless triggered by reliving the experience.

- I chose to do an MVP without annotation capabilities because I find these features of secondary value.

Should Facebook have a Travel Feature?

Question

Facebook is considering getting into the Travel industry. Should they do it?

What is this question about?

This product management interview question is a typical market entry question. The interview question tests whether you have a structured way of analyzing market entry questions using strategy fundamentals.

What is the interviewer looking for?

The interviewer is evaluating you on the following:

- Can you apply basic strategy principles? (e.g., Porter's Five Forces, the 5 Cs)
- Do you structure your analysis in a clear and logical manner?
- Can you synthesize your analysis and objectively support your final decision?
- Can you backup your decision with estimates?
- How to structure your answer?

- **Where to Play:** Identify where in the travel-industry value chain Facebook can provide a solution, given its core mission and synergies with a particular service.
- **How to Win:** After choosing where to play (services in the value chain where Facebook can add value), run a competitive and customer analysis. What is the competition offering? Determine if Facebook possesses a business or technology capability that would differentiate it and provide a competitive advantage.
- **Core Capabilities:** Determine which core capabilities would Facebook require to compete in the services chosen. Evaluate the feasibility of acquiring missing capabilities. Examples of core capabilities are technology, market knowledge, customer knowledge, partnerships, sales knowledge, relationships with suppliers, and others.
- **Size the Opportunity:** Estimate the size of the market for the service chosen and its growth rate. Is it worth investing in entering this market?
- **Recommendation:** Give a final recommendation based on Facebook's strengths in core capabilities and the size of the opportunity.

Answer Example

INTERVIEWEE: There are multiple companies in the travel industry that provide different services such as travel recommendation sites and reservation sites for airline tickets, tourist packages, accommodation reservations and others. The first thing I would do is decide where in the value chain Facebook could play. Second, determine how Facebook can win, in other words, does Facebook have a competitive advantage to make it a market leader in the chosen space. Third, decide if Facebook has the core capabilities to play in this space or does it need to acquire them. As a result of the last two steps, competitive advantage and core capabilities, I may narrow down the services that Facebook is best fit to target. Finally, I will evaluate whether it is worth entering this market based on the previous analysis, opportunity size, and risk.

How does that sound?

INTERVIEWER: It sounds promising, please go ahead.

INTERVIEWEE: Okay, let's start with describing the travel industry value chain to get a lay of the land:

I will begin by listing some of the existing mainstream services in the travel industry value chain.

- Recommendation sites, like TripAdvisor, where people go to share and find information about other people's travel experience to inform their decisions about where to go and what to do.
- Tourist operators, like Viator, offer information and reservation services for tourist packages.
- Online Travel Agencies (OTAs): OTAs like Priceline and Orbit, where people make direct reservations for air tickets, hotels, and car rentals.
- Online hospitality businesses, like Airbnb, for travelers to plan and reserve their accommodations.
- Work travel online reservation systems, like Concur, provide a booking solution system for enterprises to use internally to manage business travel. Enterprise employees make work travel arrangements that go through an approval process that helps companies track travel expenses.

I may have left some services out, but these are the ones that seem more prevalent in the industry. Is there one you think I should include?

INTERVIEWER: No, this is a good list. Please continue.

Where to Play

INTERVIEWEE: Okay, so the first part of my analysis is to determine where Facebook can play in this value chain. To answer this, I would look at two things: a fit with Facebook's mission as a company and synergies between the travel industry and Facebook's core products. Finding a fit with the company's mission is important to align management with the new market objectives. And, technology and market synergies between the travel industry and Facebook will make it easier to attain market leadership faster.

Given Facebook's mission statement, "to give people the power to build community and bring the world closer together," I think the first four travel services fit within this mission. A traveler's site, like Travelocity, helps people in the world build community and come together; tour operators and online travel agencies help people travel to new countries to experience other cultures; and the hospitality businesses, like Airbnb, help people come together by sharing the way they live. However, I do not see how Work travel businesses, like Concur, fit within this mission. Solutions, like Concur, target enterprises to make travel logistics more efficient which does not align with Facebook's mission.

Regarding technology and market synergies, these four services align with Facebook's technologies. These sites already try to personalize and create communities to enhance their customers' experience. This is an area where Facebook has technology advantage and user knowledge. So, I would pick these services as possible places to play. Do you have any questions so far?

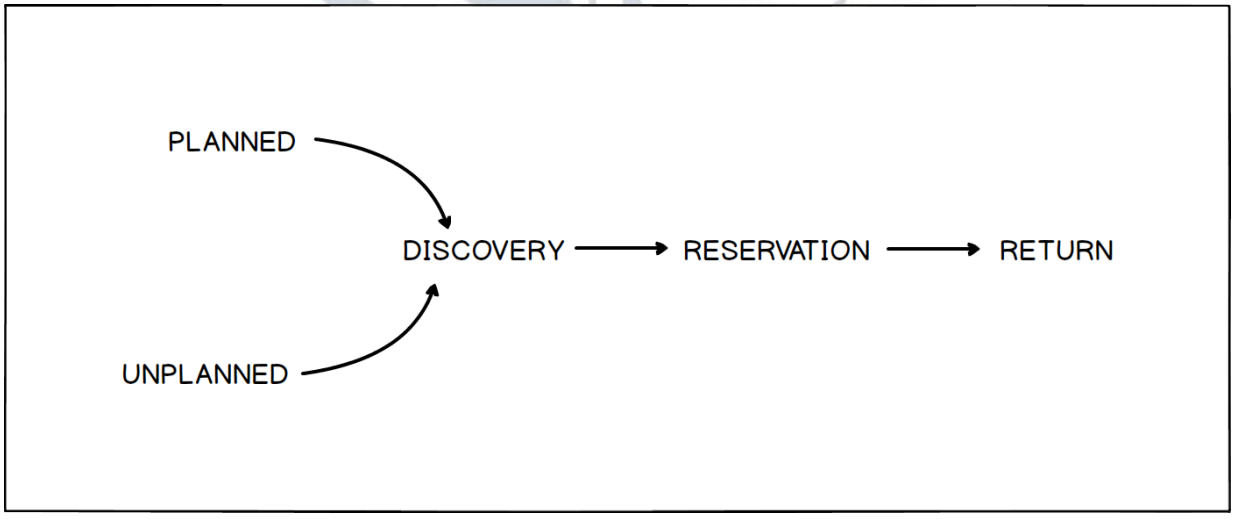
INTERVIEWER: No, that sounds reasonable.

How to Win

INTERVIEWEE: Having picked where to play, I will continue with the next step, analyzing how Facebook can win.

To guide my analysis of how Facebook can win, I would like to use a customer journey map as a way to identify what users do and their pain points. A customer journey map will help determine where Facebook can provide a differentiated experience to customers in order to gain market share from the competition.

Here is a customer journey map that many people using these services follow:



Customer Journey Map of a traveler

Pleasure travel drives a large portion of the travel industry; in fact more than 50% of the industry revenue is due to pleasure travel, as far as I know. Since I discarded work travel as a compatible market for Facebook, I am going to focus on pleasure travel. People plan ahead before deciding to travel on the spur of the moment. So the decision to travel

is *planned* or *unplanned*. In either case, if the decision is planned or unplanned they enter the discovery process and go to travel sites like TripAdvisor to get recommendations for places to visit. After browsing recommendations and deciding where to go, they proceed to make a reservation. Travelers might do the following: use an OTA to reserve flight tickets, visit Viator to reserve local tourist activities, and book accommodations using Airbnb. At the end of their travel experience, some travelers may return to these sites the next time they travel.

I think Facebook can transform this customer journey into a much more personalized experience for the user. Facebook can do this by leveraging its proprietary technologies in social networking, mobile apps, and VR. And, with a daily world audience of to 1.5B+, Facebook has a mass market to jump-start any initiative.

I will go through a few scenarios to illustrate what I mean.

Before Discovery

For travelers that have not decided yet to travel, Facebook is in a pretty good position to raise external triggers to lead them to travel. Its 1.5B+ daily users read their feeds several times a day. Triggers can be raised via friends' posts about their traveling experience, or posts/videos of celebrities that are traveling, or reminders of the great time they had in Rome last year.

Doing the same with Instagram will likely have a higher impact because of its photo-centered experience. Facebook could be a key lead generator to travel reservations by using its social platform, mobile apps, and massive audience.

Discovery

Facebook can curate photos from its network to show possible destinations once users decide to travel. Facebook can even use its VR technology to create an almost real-world experience of the final destination. Stories from friends about their travel experience could be used to enhance the impact of the photos. While recommendation sites like TripAdvisor also provide recommendations, Facebook has the advantage of having a larger pool of users to tap photos and stories from; in addition to including stories from friends that are likely to be influential.

Reservation

In the reservation process, the user books a variety of services such as airline tickets, car rentals, tour packages, hotels or rooms. This part of the process is transactional in nature so there is less opportunity for personalizing the experience. Still, Facebook could enhance this part by providing up-to-the-minute information about the chosen destination, hotels or activities that the users decided to book. If people book local activities, Facebook could enable users to connect with others that share the same activities, ahead of the trip.

Return

The next time they travel, people may go through the same process of discovery and reservation. Facebook can use its platform to get users to come back for their next travel reservations by reminding them of their past trips using anniversary notes and personalized pages. Users could create personalized pages to show photos and stories of their trips.

These are just a few examples of how Facebook can use its social platform and technologies to exert competitive advantage over other players in this space.

Now, let's move to determining if Facebook has the core capabilities needed to support these services in this customer journey.

Core Capabilities

First, I will list the core capabilities Facebook has and does not have, and how this can affect the decision to play in the travel industry.

FACEBOOK HAS	FACEBOOK DOES NOT HAVE
For discovery:	**For discovery:**
A social platform to enable users to be inspired by other users' travel experiences	Customer knowledge — what are the travelers pain points and goals?
Mobile apps (e.g., Instagram, Messenger, WhatsApp) to reach users across platforms and contexts	**For reservation:**
Fine granular personal data about users to personalize experiences	A booking system — no integration with airlines/agents GDS (Global Distribution System)
Massive potential market (1.5B+ DAU)	A payment system that integrates with airlines
	Reservation technologies that OTAs own (like price comparisons, notifications of low prices, etc.)
	Partnerships (e.g., with possibly OTAs, tourism agencies)
	Suppliers (e.g., airlines, hotels)

Listing the core capabilities Facebook has and does not have

Okay. Let's start with what Facebook does have. It has the technology and data to enhance the discovery experience of travelers. Facebook's social feed and other properties like Instagram, Messenger, and Whatsapp can be used to reach a vast audience to get them into the travel funnel through triggers. This is a unique advantage Facebook has over competitors. Facebook and Instagram have become a daily habit of its 1.5B+ users, an audience that TripAdvisor or OTAs do not have.

Additionally, Facebook's competitors do not have granular personal data about their users. Facebook does and can use that data to provide a very personalized discovery experience.

For example, using travel photos or posts that users liked, shared or commented on, Facebook can infer where users want to travel.

However, Facebook lacks core capabilities in other areas. For example, OTAs have been in the travel business for years. They understand traveler needs and wants when booking a flight or hotel. Facebook does not have this type of travelers knowledge. It is something they would need to acquire by hiring people from the industry.

When it comes to the reservation part of the customer journey, Facebook lacks the technologies to complete the reservation. For example, Facebook does not have a GDS booking system or a payment system that connects to airlines. Facebook also lacks the technologies that OTAs have been optimizing for years to provide a better experience for the traveler when booking. Some examples are price comparisons between different airlines, letting users pick their seats, and letting users subscribe to alerts for special deals on flights or hotel rooms. Finally, Facebook does not have partnerships or relationships with suppliers, such as airlines and others.

So looking holistically at the core capabilities, Facebook would be best positioned to enter the travel industry in the discovery part of the business, where it has most of the core capabilities. Facebook could provide a dedicated space on its platform, via an app or page, where users can research travel planning. And ad clicks for airlines tickets, hotel reservations, activity bookings, or tourist packages would generate revenue.

In the long term, as Facebook learns more about the travel industry, and users see Facebook as a reliable place to do travel planning, Facebook could enter the Reservation space and compete directly with OTAs. That would require a lot more capital investment, as well as technology integration with GDS and payment systems. So it is a riskier move.

INTERVIEWER: So how would you evaluate the opportunity of the discovery market?

INTERVIEWEE: Yes, that is the next step. I would evaluate the opportunity of the discovery market by estimating the revenue Facebook can make from travel ads.

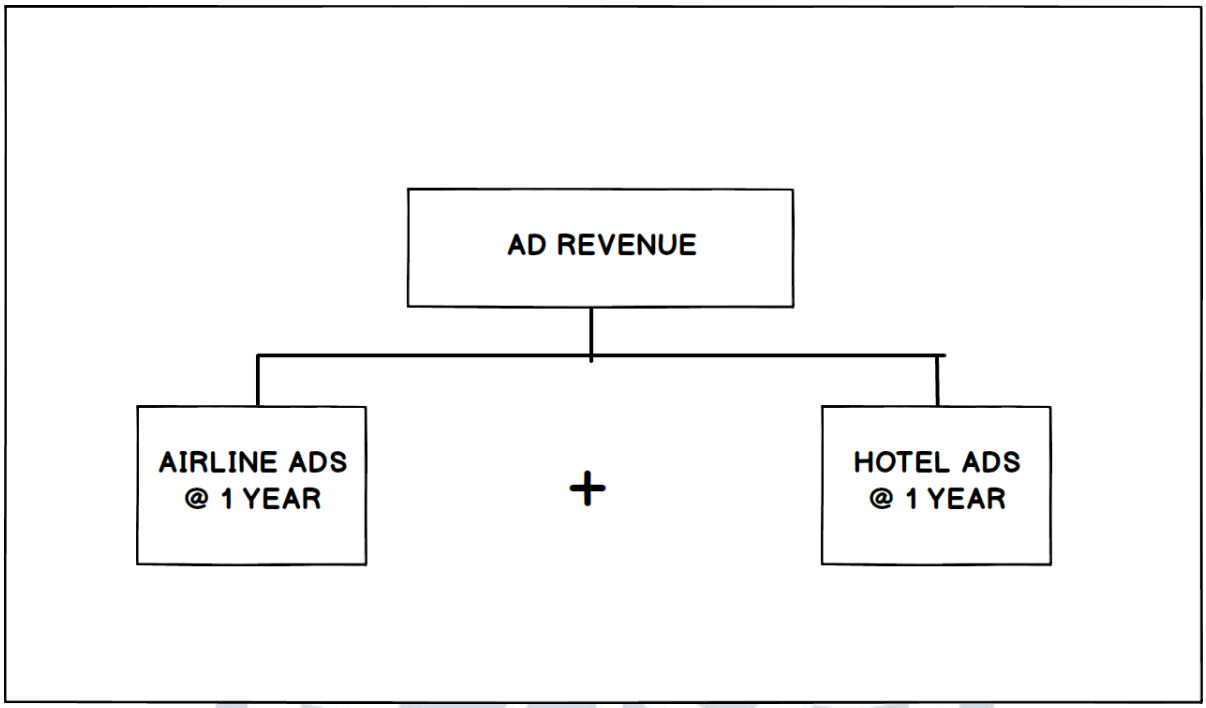
INTERVIEWER: Okay. Go on, please.

INTERVIEWEE: For starters, I would only consider the US+Canada and Europe because those are the world areas where people have the income to travel for leisure, and marketers are likely to target these markets. Although Chinese people are frequent travelers, Facebook, like other American internet companies, have been blocked from China, so I am going to leave China out.

So assuming Facebook has a dedicated page or an app for travelers, ad revenue will come from airline ticket ads, hotel ads, activity ads, or package tour ads. But, to simplify this exercise, I will consider only ticket and hotel ads, which today are the most prominent. Does that seem reasonable to you?

INTERVIEWER: Sure.

INTERVIEWEE: *(The interviewee uses a whiteboard or notepad to draw an estimation tree that will help explain the assumptions and calculations. At the root of the tree is "ad revenue" and the first level of branches are "revenue from airline ads" and "revenue from hotel ads.")*

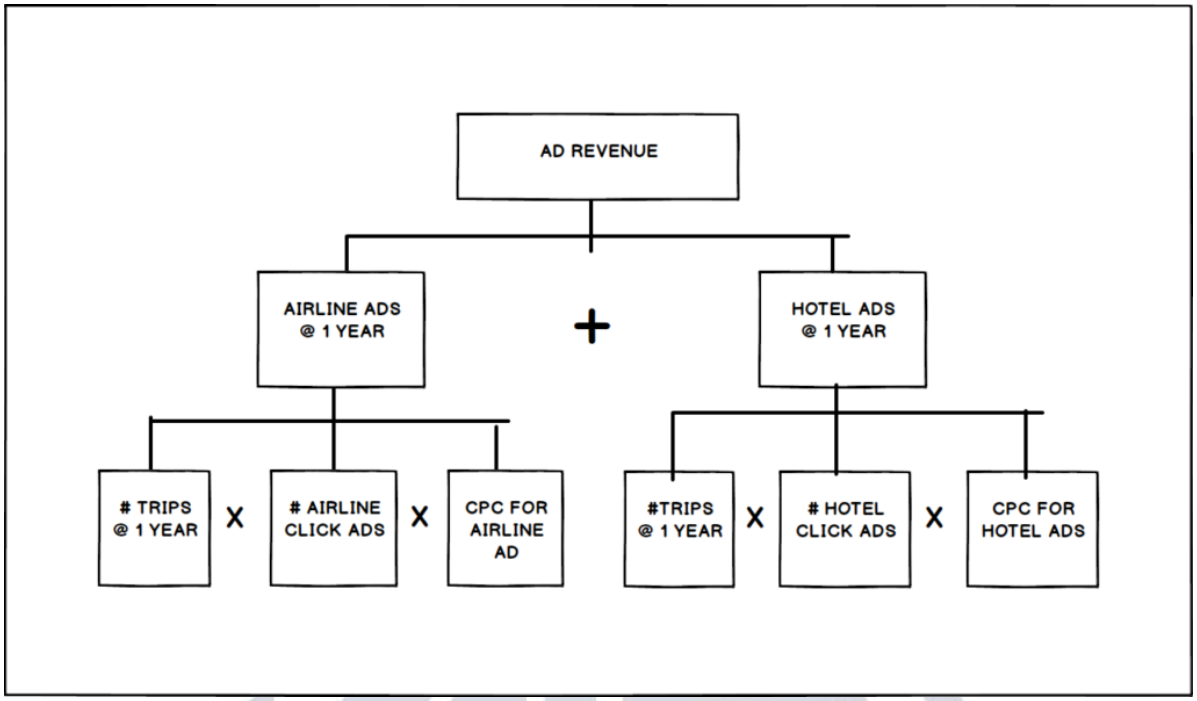


Estimation Tree first level operation: airline ads revenue and hotel ads revenue
To estimate ad revenue for a year, I will first estimate ad revenue from airline ads and then from hotel ads. Then I will add these two ad revenue sources to compute total ad revenue.

INTERVIEWER: Okay, that makes sense. Please continue.

INTERVIEWEE: *(The interviewee draws the second level branches and then explains.)*

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Estimation Tree second level operation: decompose airline ads and hotel ads into #trips, #ads and cost

Revenue for airline ads can be estimated by multiplying the average number of trips per year Facebook users in US+Canada and Europe take, times the number of airline ads they would click on, times cost per click (CPC) per airline ad.

Revenue from hotel ads can be estimated by multiplying the average number of trips per year Facebook users in US+Canada and Europe take, times the number of hotel ads they click on, times the CPC per hotel ad.

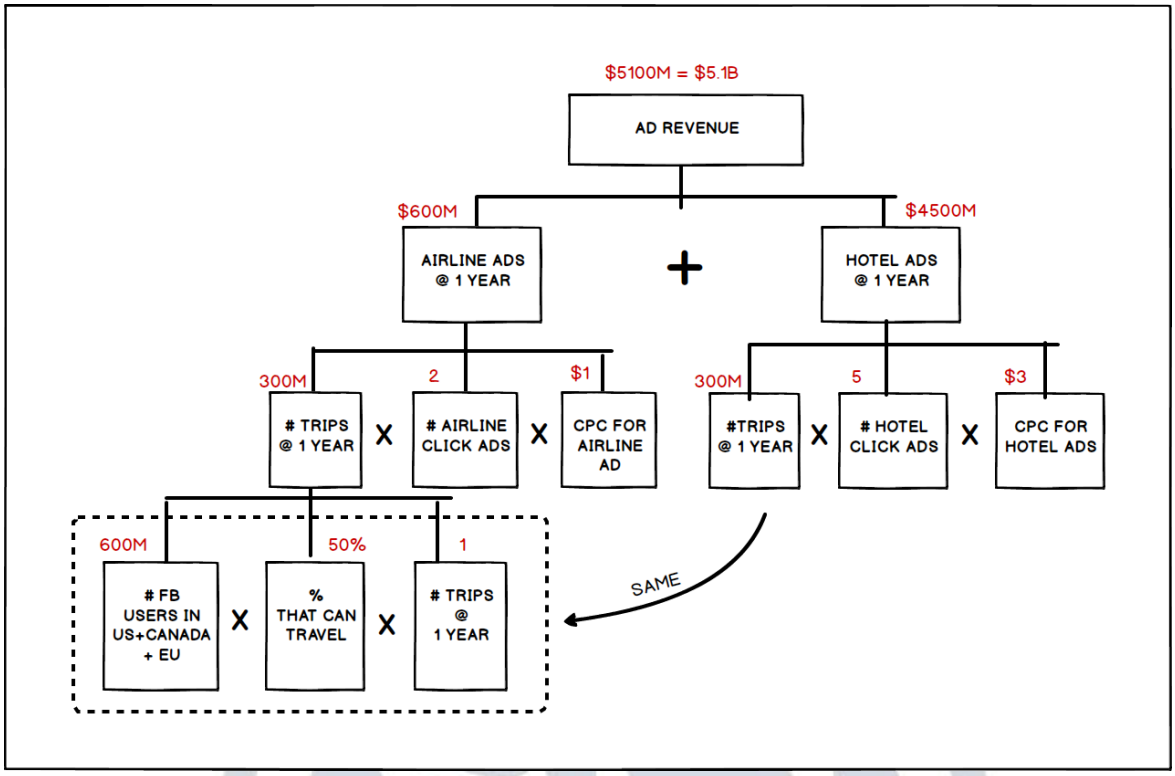
INTERVIEWER: So are you saying that every Facebook user in US+Canada and Europe that travels will go through the Facebook travel services?

INTERVIEWEE: Yes, this is assuming 100% penetration because I want to size the potential market first. I will bring down this penetration to 10% at the end to get a more realistic estimate. Not everyone will use the Facebook services at launch time, but I think that 10% is a realistic target.

INTERVIEWER: Ok.

INTERVIEWEE: Now let's see how we can calculate the number of trips per year that Facebook users in US+Canada and Europe take.

(The interviewee draws and starts explaining the calculations starting from the leaves of the tree.)



Complete estimation tree

I can estimate this number by multiplying the number of Facebook users in these two continents, times the percentage of those users that have the income to travel, times the average number of trips per year they take. I know that the number of Facebook users in the US+Canada is about 250M. And, I am going to guess that in Europe it is higher, given the larger population. My estimate would be about 350M, adding to a total of 600M Facebook users on both continents.

(The interviewee writes down 600M on the top left-hand corner of the box that says FB users in US+Canada and Europe. The interviewee continues calculating the numbers for the rest of the boxes. This process helps the interviewer understand how the numbers are calculated and aggregated at each level of the tree. This leaves no doubt as to how the interviewee is arriving at the final estimate.)

My sense tells me that a person in the US or Canada needs to make at least $50K a year to afford a leisure trip. That is about 50% of the working population in the US, and I will assume the same for Canada, where the standard of living is similar. I believe incomes are higher per capita in Europe, but to be on the conservative side, I will assume 50% for this population too. Now, I take about one trip a year for vacation, and I think this is the same for most people with an average income. So multiplying these three numbers we have, 600M users X 0.5 X 1, which is 300M trips per year.

Going up one level in our tree, we can calculate the revenue for airline ads in one year. It is equal to 300M trips/year X the number of airline ads a user clicks X CPC for an airline ad. The number of airline ads I click when making reservations is probably 2, and I know that sites like Kayak charge a CPC of $1 for airline ads. Replacing these numbers then, the revenue for airline ads per year is 300M trips/year X 2 ads X $1 = $600M.

Okay, let's move to the right side of our tree and calculate the revenue from hotel ads in one year. From our previous calculations, we estimated that the number of trips per year to be 300M. When I research for hotels or accommodations, I usually click on some hotel ads to check them out, maybe five. Now, I know that CPCs for hotel ads are more expensive than for airline ads, about three times, so I am going to assume that CPC for hotel ads is $3. So multiplying these three numbers results in 300M trips/year X 5 ads X $3 = $4500M for hotel ads per year.

Adding the ad revenue for airlines and for hotels results in $600M + $4500M = $5.1B per year.

$5.1B per year is the total potential revenue assuming a 100% penetration. But since Facebook would compete with all the other players in this market, I would start with a 10% penetration assumption, as I suggested earlier. So 0.1 X $5.1B is $500M for the first year.

INTERVIEWER: How can you check that your numbers are not too off?

INTERVIEWEE: Well, I know that the big airlines make on average around $40B a year, and they do not spend more than 1% on advertising. So if we just consider airline advertising and we divide my estimate by the total revenue major airlines make in a year, that should be under 1%. Let's check. There are about five major airlines in the US that make about $40B, and I would imagine there is half that number in Europe, say 2. Then dividing our estimate of $600M by $40B * 6 = 600M/240B, is about 0.0025 or 0.3%. This is much lower than 1%, but that is expected, considering that our estimate only includes Facebook users and not the entire population in US+Canada and Europe that travels.

INTERVIEWER: Okay, continue.

INTERVIEWEE: Sure. $500M is not a huge market when compared to the $40B+ that Facebook makes in total ad revenue. And, it is not a fast growing market either. The travel industry is an established industry, its yearly growth is in the lower digits, around 5%. However, on the bright side, it would not take much capital for Facebook to develop features to enter this market. Facebook already has the data and technology and only has to invest in development and maintenance which requires just labor.

INTERVIEWER: So what is your recommendation?

Recommendation

INTERVIEWEE: Well, I think the bigger opportunity is in reservation services. Transaction fees for airline tickets, hotel bookings, activity bookings, and other travel reservations can generate higher revenue. Doing a rough estimate, we estimated 300M trips a year, so charging a fee of 10% for an average airline ticket of $500 results in 300M x 0.1 x $500 = $60000M or $60B, take 10%, that's $6B, just in airline tickets. I have not included hotels and activities reservations.

However, given the capital investment, competition, and lack of market knowledge in the reservation business makes entering this business risky in the short term.

So, my recommendation would be to implement the dedicated page or application feature in the short-term, with two main objectives, to acquire travelers on Facebook and learn about their needs deeply. Once Facebook acquires the travel segment among its users and obtains deep knowledge about their needs, Facebook can more confidently acquire the technologies it needs to perform the transactional part of the reservations phase. In summary, I am recommending a short-term and long-term strategy. The short-term strategy focuses on acquiring customers and learning. And, the long-term strategy focuses on using those learnings to successfully convert customers on the more riskier part of this market, reservations.

INTERVIEWER: Okay, that sounds like a clear strategy. Thank you.

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How would you find the cause of a 15% drop in Facebook Groups usage?

What is this question about?
The interviewer is testing to see if you have an organized and logical approach to solving problems. In this problem-solving interview question, there is a situation where data could be pointing to a problem or outcomes that conflict with each other. You are asked to discover what the causes are and explain how to solve the problem.
What is the interviewer looking for?
The interviewer is evaluating you on the following:

- What is your thought process when diagnosing and solving a business problem?
- Are you logical?
- Are you exhaustive in listing the possible causes?
- Are you articulate in your delivery or do you tend to ramble?

How to structure your answer?

- **Clarify.** Ask for clarification if you are unclear about what the interviewer is asking.
- **List high-level reasons.** Start by listing the high-level reasons, components, or user behaviors that are causing the problem.
- **Gather context information.** Ask questions to understand the context of the problem. For example, is the problem regional? Is the problem a one-time event or progressive?
- **Discard issues outside of scope.** If the answers to questions about context are outside the scope of Facebook Groups features, discard them.
- **Establish a theory of probable cause.** For each of the high-level reasons, components or behaviors you mentioned, recursively ask yourself why are these problems occurring, and list possible causes.
- **Explain how to test your theory and fix the problem.** Describe how you would test each probable cause. If possible, explain how you would resolve the problem.

Answer Example
Clarify
INTERVIEWEE: By usage, do you mean the creation of groups or user engagement or both?
INTERVIEWER: Strictly engagement.
INTERVIEWEE: Okay, how do you measure engagement? Is it by the number of posts created, the number of replies to posts, the number of reactions to posts, or the number of shared posts?
INTERVIEWER: All of those things. The total count of all those activities, or total engagement for Facebook Groups, has declined by 15%.
List High-level Reasons
INTERVIEWEE: Okay, at a very high-level the decline in engagement is due to:

1. Users creating, replying, reacting, and sharing fewer posts on Facebook Groups, and/or
2. Users going to an alternative feature.

To start diagnosing the cause of these behaviors, I would begin by asking the following questions to understand the context in which the decline in engagement was detected.
Gather Context Information
TIME: Is the decline in engagement a one-time event or has it happened progressively? If it is a one-time thing, then it is possible a technology glitch caused the

problem, such as a downtime in the services that support Facebook Groups. Therefore, I would ask if there were technical problems that coincided. If the decline in engagement is progressive, then the cause is still there, and we can dig deeper into data, which I will go through later.

REGION: Is the decline happening in an isolated region? If this is true, the problem might be related to a country's regulations or a competitive product in that region. For example, if there is a new competitive social-group product that is more suited to the social mores of the region than Facebook, I would do some ethnographic studies on that population to understand their needs better; and I would consider developing features for those needs.

PLATFORMS: Is the decline happening on specific platforms, like iOS, Android or others? If so, I would compare the drop of Facebook Groups engagement on each platform with engagement across all platforms. As long as the overall engagement metric is steady, I would not worry much about the decline of engagement on a particular platform. Users could be, for example, switching mobile platforms and still using Facebook Groups.

OTHER FEATURES: Is the decline in engagement happening in other Facebook features besides Facebook Groups? If so, then there is a much bigger problem, and we would need to look at the overall engagement of the entire platform.

COMPETITORS: Is the decline in engagement also happening in competitive products? This data is difficult to come by, but if this were true, then the cause is also affecting other companies. For example, the decline in engagement could be a PR problem with privacy perceptions about social networks. This kind of problem would be out of scope for a product team and best handled by a corporate management team.

Establish a Theory of Probable Cause

To continue with a more in-depth analysis, let's assume that the engagement is progressive, and none of the other situations are happening. Meaning, the problem is not regional, not platform-specific, and no additional Facebook feature or competitor is exhibiting the same decline in engagement. I would continue diagnosing what could cause users to behave in the two ways I described:

1. Creating, replying, reacting, and sharing fewer posts,
2. And, going to an alternative social-group feature.

Test Your Theories and Fix the Problem

Concerning creating, replying, reacting, and sharing fewer posts, I would ask the following questions:

- **Is there a problem with the creation of posts?** Maybe posts are created but not displayed on group members' feeds. If this were the case, it could explain why users are not engaging. However, I think a decline in engagement would have been drastic and not progressive. So, the probability of this being the cause of a progressive decline in engagement is small.
- **Are the notifications of new group posts working correctly?** Perhaps group members are not engaged because they are not getting notifications about new posts. Like the previous case, I think this malfunction would cause a drastic decline, not a progressive one. So the probability of this being the cause is small.
- **Is there a problem with the UI for replying, reacting or sharing of posts on Facebook Group pages?** If this were the case, I think users and the Q&A team would notice right away, and fix it. So, I do not believe this would be the cause of a progressive decline in engagement.

- **Has there been an increase in spam posts on Facebook Groups?** Spam could cause users to leave groups, and be a possible cause of a progressive decline in engagement. I would investigate this further and ask my engineering team to check whether there are signs of cyber attacks and find ways to increase protection against those attacks.

In relation to users going to an alternative social-group feature, I would ask the following questions:

- Is there a new Facebook feature that is cannibalizing Facebook Groups? If there is a new feature similar to Facebook Groups, then I would look for two things: 1) What is the proportion of users of the new feature that are also users of Facebook Groups? And, 2) What is the percentage of users that switched that have exhibited a significant decline in engagement with Facebook Groups since the problem started? Answers to these questions would determine if there has been an exodus of active Facebook Group users to the new feature. If this were the case, it does not mean bad news. It is possible that the new feature brings more revenue to Facebook and adds more value to users. So, I would get together with all stakeholders to assess whether it makes sense to keep Facebook Groups.
- Is there a competitor of Facebook Groups that has increased its engagement numbers while Facebook Groups' engagement decreased? I would imagine this internal data is challenging to get, but if it became public knowledge, I would find out what the competitor did differently. Then I would implement these new ideas on a test drive with a controlled group of users. If the new features increased engagement, I would implement them.

In summary, my approach to finding the cause of lower user engagement of Facebook groups is as follows: I started by understanding the context so that I could discard issues related to regions, specific platforms, side effects of other Facebook features, and marketing problems. Then, I dug deeper to identify what could affect user behavior to stop them from engaging. Of the six possible causes, I think the most plausible ones are: **an increase in spam posts**, **cannibalization by another Facebook feature**, and **an external new competitive feature**. I would test each of these probable causes to identify the exact source and then fix it.

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How would you design a Facebook home screen?

What is this question about?

This product design interview question tests whether you understand the process of going from customer needs to product development. This process involves determining who the customer is, understanding what they want to accomplish, defining multiple use case scenarios, prioritizing what to build and having good business acumen.

What is the interviewer looking for?

The interviewer is evaluating you on the following:

- Can you empathize with customers? Can you articulate their needs, feelings, and expectations?
- Are you able to provide multiple and diverse use case scenarios?
- Is your answer structured and logical, or do you ramble?
- Do you go beyond generalities in your solutions and provide detailed descriptions?
- Can you provide ideas that no other candidate has mentioned?
- Are you confident and sound credible? Would engineers and other team members follow your lead?

How to structure your answer?

To come up with innovative design ideas, you need to understand what the user wants to do relative to the product you are building. A "job" refers to the progress your user wants to achieve in a particular circumstance. If you define what this means for your user, you will gain clarity on what the key goal of your product should be. This should be the first step.

The next step is to design innovative solutions to help users achieve their goals. To do this use one of the following two techniques: create a customer journey map or list use cases. We will use a customer journey map. We chose this technique because, in the case of a phone home screen, there is a path a user goes through to achieve their goal. Describing the customer's journey will help reveal what the user wants or expects to achieve at a particular stage on their journey. And, this revelation will help spark ideas for new features.

Here is a structure for working on this question using the customer journey technique:

- Ask clarifying questions.
- Identify user types and decide if you need to prioritize one or more.
- Define the "job" to be done for the target user.
- Describe the customer's journey to uncover user needs.
- List features that address those needs.
- Prioritize features.

- Brainstorm solutions.
- Evaluate the trade-offs of each solution.
- Summarize: state the problem, what the solution does, and why it is the best solution.

Answer Example

INTERVIEWEE: I believe Facebook launched a home screen feature in 2013. Am I correct?

INTERVIEWER: Yes, but it was not successful.

INTERVIEWEE: Why?

INTERVIEWER: The feature was called Facebook Home. It failed because Home prioritized displaying Facebook content when users' main pain point was to access their favorite apps. The feature covered the device's screen with a feed of Facebook photos, notifications, and posts. It was a beautiful interface, but it missed the crucial fact that users wanted direct access to their favorite apps and widgets. Facebook Home did have an app launcher that provided access to apps, but it took too many clicks to access them.

INTERVIEWEE: So, why try it again?

INTERVIEWER: Well, if we can design a home screen that appeals to the majority of users, we think we can increase engagement across many Facebook properties.

INTERVIEWEE: So, is the business goal to increase engagement of Facebook properties?

INTERVIEWER: Yes.

INTERVIEWEE: Is there any particular region that this new home screen will be targeted to?

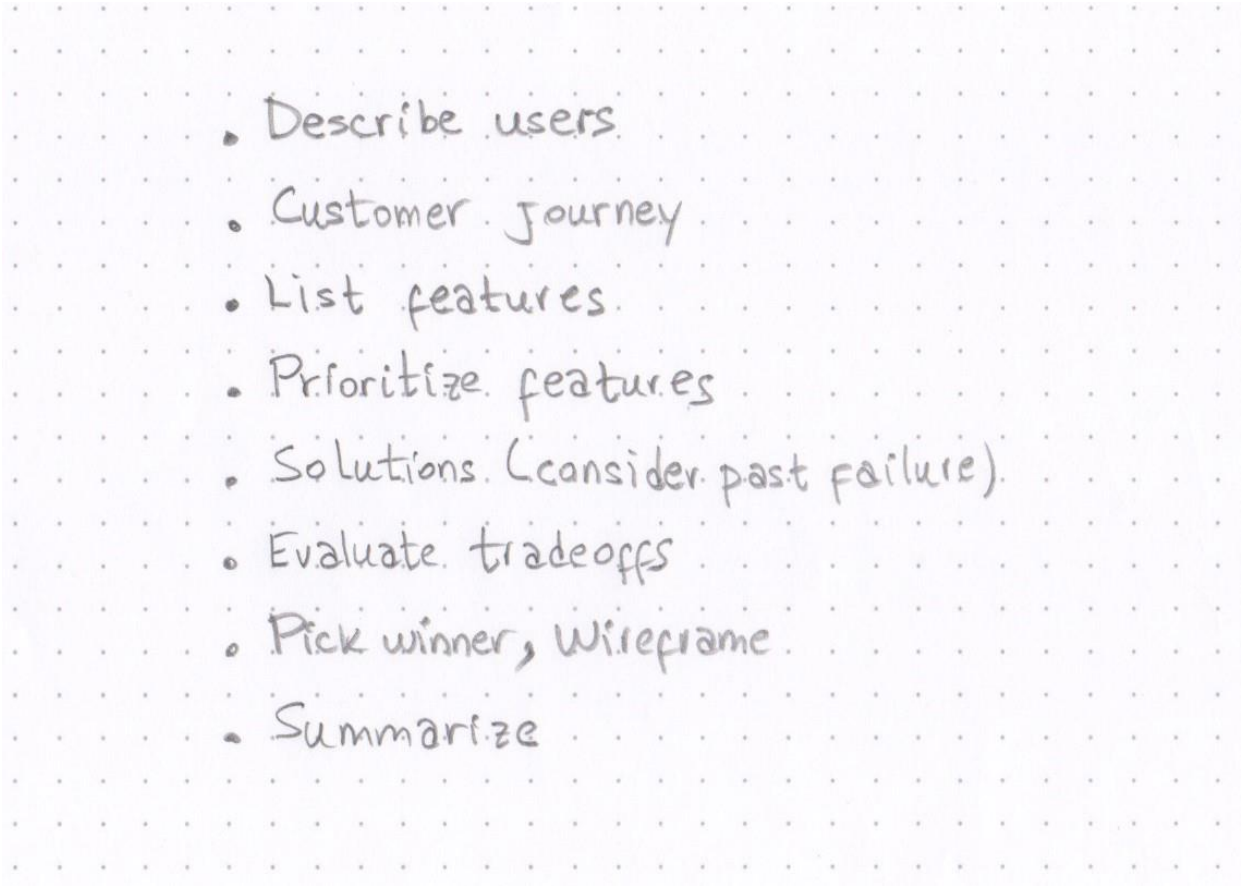
INTERVIEWER: The US and Europe. They represent a large percentage of Facebook users.

INTERVIEWEE: Okay, could I take a moment to think about how I will approach this question?

INTERVIEWER: Yes, please do.

(The interviewee takes a minute to write down how she will structure her analysis.)

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Interviewee's notes on how to structure answer

INTERVIEWEE: Okay, so Facebook Home failed because the design did not take into account a key user need: **to have fast access to apps that will help them get a task done**. Granted Facebook is an immensely popular app that people access several times a day. But, it is not the only app they use. So, I would like to take a different approach and prioritize the user's needs relative to the use of a home screen. I will start by analyzing the customer journey towards discovering and using apps. Along the journey, I will identify the user's needs, list some features to address those needs, and then brainstorm some solutions. At this stage, I will consider ways to include features within the home screen to get users back into Facebook properties. If users find the new home screen valuable, they will start using it frequently. And, as users are exposed to other Facebook properties within the home screen, they will likely use them more frequently. How does this sound?

INTERVIEWER: Sounds good, please proceed.

INTERVIEWEE: Facebook users come from all demographics and lifestyles. Some users are more tech-savvy and may like to customize the presentation and organization of their app icons. But, I think most users do not want to deal with customizations. In the end, the main reason they are using apps is to get something done. And, the faster they can access the app they need, the faster they will get a job done. Therefore, I think the goal of the new Facebook home screen feature should be **to help users get things done faster**. In the context of the home screen, this can mean finding or opening an app faster, or something else.

Let's walk through a typical user journey to find and use apps to get things done. This process will provide insights into what actions need to be sped up.

1. A user presses the home button and sees the apps screen.
2. The user looks for a particular app.
3. The user cannot visually identify the app and resorts to search using a text-based search or voice search.
4. The user finds the app icon and touches it to open.
5. The user may want to switch to another app.

I think this workflow can be improved to help users get things done faster. Here are some ideas for features:

Present the most appropriate app icons to the user when needed.
With gazillion apps available, it is increasingly difficult to find an app visually. Taking the time to organize apps into folders is not a good solution because it is hard to remember which folder an app is in. Plus, it is unlikely users enjoy the chore of organizing apps.

Open the most appropriate app automatically.
Instead of just presenting the most appropriate apps for the user, why not open them automatically?

A quicker way of switching from app to app.
Create a way for the user to switch to another app, without having to go through the process of finding an app again.

To decide on which feature to prioritize for development, I will now compare them against four attributes:

1. How important is the user's pain point the feature solves,
2. How innovative is the feature relative to competitors' features,
3. How much effort will it take to build,
4. And how risky it is to build the feature.

To numerically prioritize the features, I will use a scoring system, from 1 to 5, to rate a feature across these attributes. I will add the positive attributes and subtract the negative ones. The positive attributes are: "importance of pain point" and how "innovative" is the feature. And, the negative attributes are: "effort" to build the feature and "risk" level.

I will make the comparison using a table.

FEATURE	IMPORTANCE OF PAIN POINT (+)	INNOVATIVE (+)	EFFORT (-)	RISK (-)	TOTAL
Present most appropriate app icons	4	1	4	3	= 5 - 7 **= -2**
Open most appropriate apps	5	4	5	5	= 9 - 10 **= -1**
Switch to another app	3	1	4	3	= 4 - 7 **= -3**

Prioritization of features across attributes

Pain Point Attribute

Let me start by rating the feature on the "importance of pain point" attribute. The feature, "open the most appropriate apps," addresses a key pain point towards achieving the goal of helping users get things done faster. Of the three, this feature provides immediate access to apps the user wants at the right time. Therefore, I give it 5 points.

The "present most appropriate app icons" feature is similar, but it only displays the app icons. It does not open the apps. So, I give it a 4.

And, the "switch to another app" feature addresses a less critical pain point, since not all users may want to switch to another app. So, I give it a 3.

Innovation Attribute

For innovation, I think the "open the most appropriate apps" feature is the most innovative of the three because it does not exist as far as I know. Therefore, I give it a 5.

The "present the most appropriate app icons" feature is something that competitors like Aviate already do. Aviate has an app launcher that presents the right app icons at the right time. The app analyzes what the user is doing, the context, and displays the most relevant apps for that context. For example, if a user is travelling, the app will show the user's Google Maps app, navigation apps, and Yelp. Therefore, I give this feature a score of 1 for innovation.

The "switch to another app" feature is something Aviate already solves. So, I give it a score of 1 too.

Effort Attribute

For Effort, I think the "open the most appropriate apps" feature takes more effort than the "present the most appropriate app icons" feature. Because it involves finding the right apps based on context and designing the UI to display opened apps, I give it 5 for effort.

The other two features: "present the most appropriate app icons" and "switch to another app" use the same algorithm to select apps as the "open the most appropriate apps" feature. But, the UI design is less demanding. So, I give both of these features a lesser score of 4.

Risk Attribute

For risk, I think the "open the most appropriate apps" feature is the riskiest to develop because it chooses which apps to open on behalf of the user. While the other two features let the user decide which app to open. In the first case, the user will blame the feature if an app is not visible right away. While in the second case, the onus is on the user to open the app. So, I give the "open the most appropriate apps" feature a 5 for risk, and 3 to the other two features.

After summing up the rows, the "open the most appropriate apps" feature ranks the highest, so I would prioritize this feature for development.

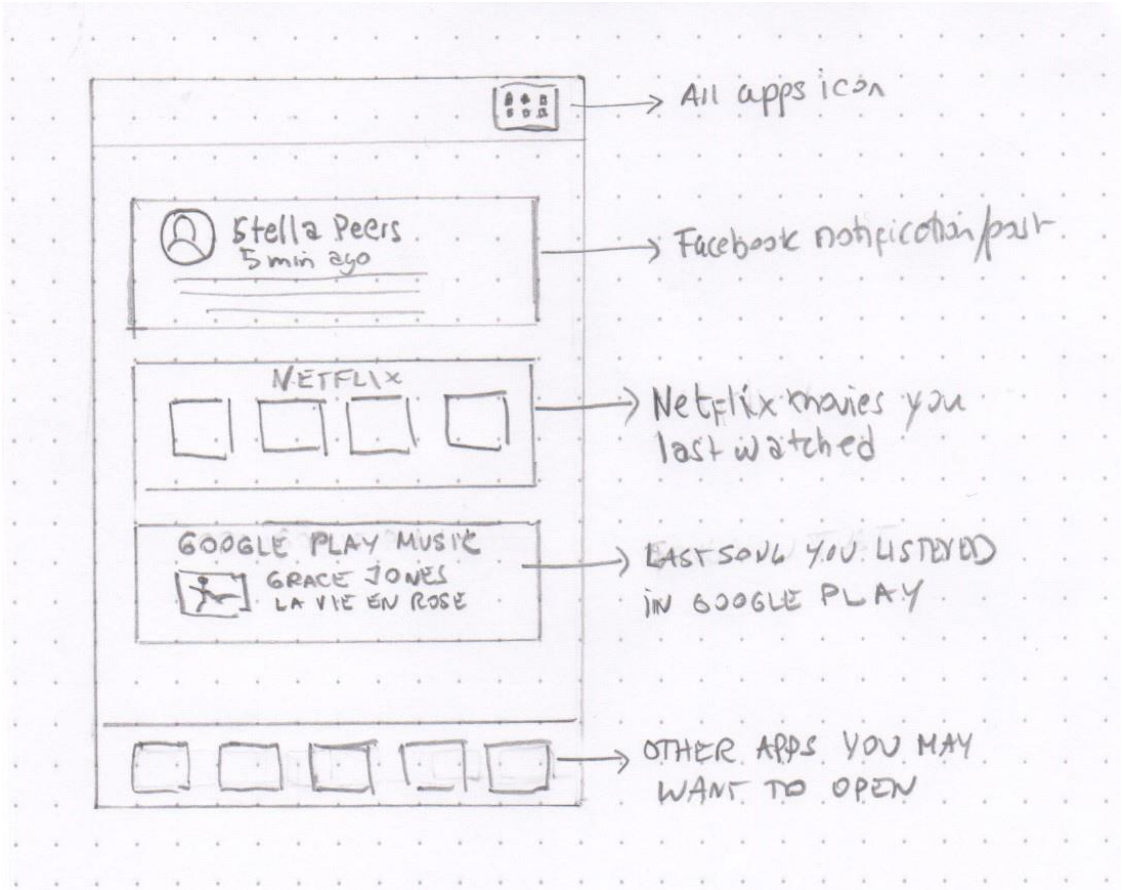
INTERVIEWER: Okay. Next, can you think of a few designs for this feature?

INTERVIEWEE: Sure, I can think of two.

The first design displays content of the most frequently used apps in a table view. The table view presents an app's partial content inside a cell and lists the apps in order of importance to the user. The user can scroll up or down and select the app they want to display in full-screen mode.

To get users to use Facebook apps, I would display these apps at the bottom of the screen in the Tab Bar. This way, users would have easy access to Facebook apps, like the regular Facebook app, Messenger, and Instagram. Also features within the regular Facebook app, like Marketplace and Live, could be displayed in the Tab Bar to enable direct access. Also, if there are Facebook posts where the user was tagged, those can be shown on the list. The post disappears after being viewed.

I think it is important to provide easy access to all apps outside the FB home screen. There are users who would still want the option to go to the apps screen to view all their apps. For these users, I would design an icon that takes users to the apps screen when touched. And, to get back to the Facebook home screen, the user could touch a back button on the app's navigation bar. Let me sketch what I have in mind.



Wireframe of new Facebook Home Screen

INTERVIEWER: Nice, thank you. Tell me, how would you choose the most appropriate apps to present?

INTERVIEWEE: I would use a supervised machine learning algorithm to classify which daily scenario the user is in, and then display the user's favorite apps for that scenario. For example, if it is morning and the user is at the gym and walking on a treadmill, this is a particular scenario. And, in this scenario, the user may have some favorite apps, like a favorite music streaming app, or a favorite video streaming app, or a favorite news app, or a favorite exercise app. Inputs to the algorithm to classify the particular context can be the time, the user's location, the user's motion, and the types of apps used by the user. Then apps used by the user in those situations can be grouped and display to the user ranked by frequency of use.

INTERVIEWER: Okay, how about your second solution?

INTERVIEWEE: This solution is inspired by the idea of the super app, like WeChat. Make the Facebook home screen the main portal to everything the user needs to get a job done. This is what WeChat does, except WeChat is a messaging platform. In addition to providing access to all apps, WeChat enables the user to interact with services and transact. Messaging capabilities are critical to enable the user to communicate back and forth with services.

So, I propose making the Facebook Home Screen a messaging-enabled feature. We would still apply the idea of displaying only the most appropriate apps and service apps to the

users, but with this solution, the user can communicate and transact. Lifestyle activities—making appointments, scheduling a service, talking to customer service, and ordering food—would be possible from the Facebook Home Screen.

INTERVIEWER: Okay, that is ambitious. Do you see any issues with these two solutions?

INTERVIEWEE: The main issue that I see with the first solution is that personalization of the apps can get in the way of the user finding new apps. This solution stifles variety, and the user can become bored of the same apps. One way to solve this would be to have a section, perhaps at the end of the table, that displays new suggested apps. And, the new apps could be picked based on which apps are trending.

An issue I see with the second solution is that competitors might view it as a monopolistic business threat. Since there are 2B+ Facebook users worldwide, Facebook has a distinct advantage over other businesses when it comes to having access to users. So, I can see how this could be a contentious issue among regulators, particularly in the US and Europe.

INTERVIEWER: Okay, those are good points. Which solution would you implement if you were to choose one?

INTERVIEWEE: Of the two solutions:

1. Context-based display of content of the rights apps at the right time and
2. A super home screen

I think the first solution achieves the immediate user goal—to get things done faster. It is less ambitious than the second solution, which means it is less risky. The second solution achieves the goal too, but it is much more ambitious and very risky. It is riskier because of possible regulatory hurdles and uncertainty about how western users would like it. Although Chinese users like using WeChat as the portal for everything, people in the western world may not react the same way. So, I would go with the first solution. It achieves the goal of helping users get things done faster and is less risky.

Would you like me to summarize my recommendations?

INTERVIEWER: Yes, please do.

INTERVIEWEE: Since the main reason for the failure of Facebook Home in 2013 seems to be that it hindered users from having fast access to their favorite apps, I proposed something different. I suggested designing a home screen that prioritizes the user's key goal: to get things done faster with their phone. The solution I recommended is a context-based solution that displays partial content of the right apps at the right time in a table view. I preferred this solution over the second one, the super home screen, because it solves the user's key pain point and is less risky. This new home screen adds great value to the user, and therefore the user is likely to use it very frequently. As a result, users will engage more frequently with the other Facebook properties displayed on the home screen, achieving the desired business goal.

How would you prioritize new product features for Facebook?

What is this question about?
Prioritizing product initiatives and product features is at the core of product management. This interview question tests whether you have a clear process of prioritization and whether you can boil down your prioritization to numerical comparisons.

What is the interviewer looking for?

- Do you have a method for prioritizing features?
- Can you formulate how to assign numerical values to different levels of priorities?
- Can you boil down your prioritization to a numerical comparison?
- Do you explain your assumptions clearly behind the values used in your formulas?
- Are you organized and structured in your analysis?

Answer Structure

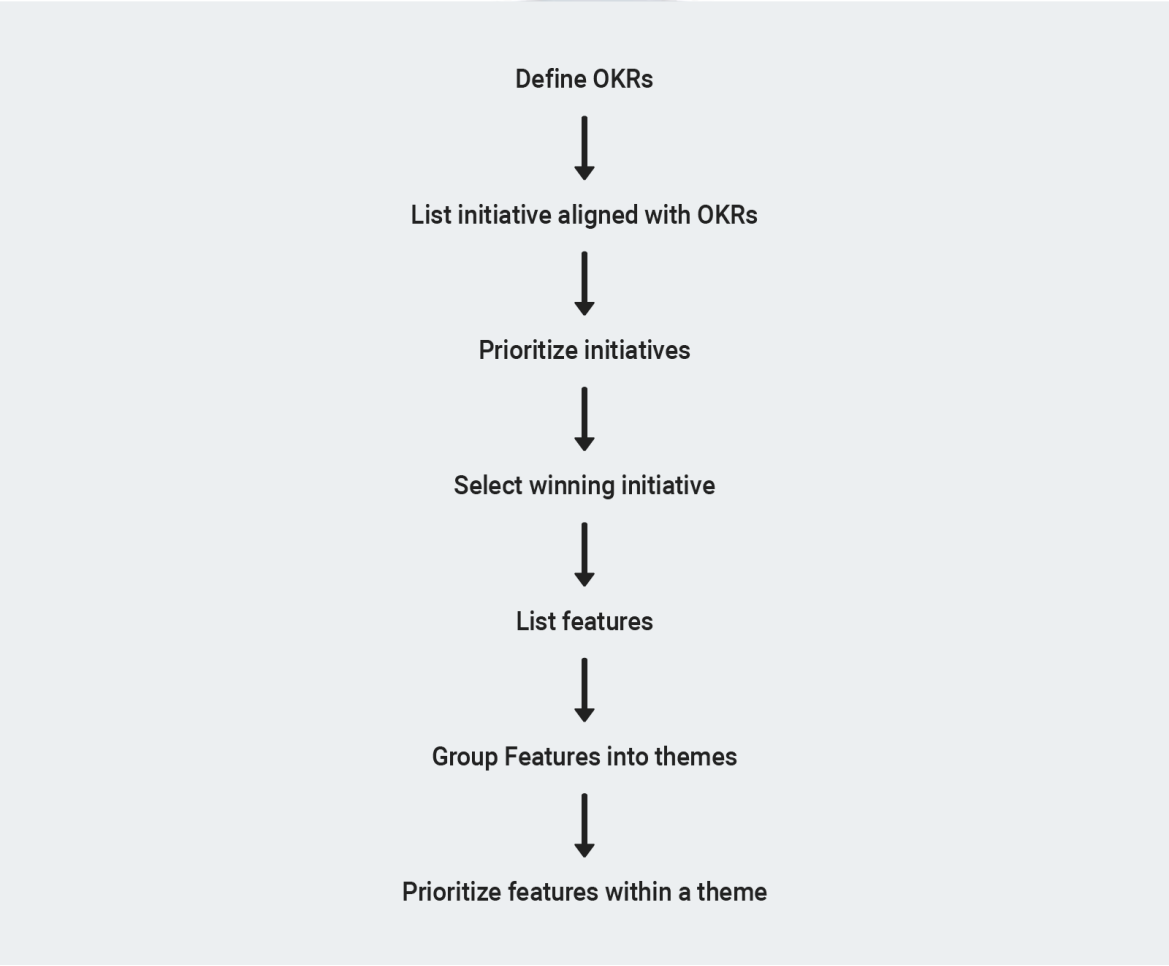
- **Begin by describing your prioritization process.** Drawing a diagram will help the interviewer follow your explanation better.
- **Apply your process to an existing product to demonstrate how it works**, or ask the interviewer to choose an example.
- **Before moving to the next step in your prioritization process**, tell the interviewer which step you just concluded and which step will be next.
- **Use a benefit vs. cost analysis or ROI** when prioritizing initiatives.
- **Explain your reasoning behind each term** when formulating equations.
- **Formulating all equations before assigning values** will keep your work more organized.
- **Create a two column table to evaluate your equations numerically.** Use the left column to write assumptions and numerical facts about the terms used in your equations. In the right column, replace your equation terms with numerical values.
- **Calculate benefit/cost (ROI) for each initiative.** Check if the winning initiative meets the result expected from the Objectives & Key Results (OKR). State which initiative is the winner and continue prioritizing features for that initiative.
- **Use a scorecard method for prioritizing features.**
- **Summarize your analysis and provide your recommendations** of what initiatives and features to build.

Answer Example

INTERVIEWEE: To prioritize new product features for Facebook, I would start with the OKRs the business wants to achieve for the period of interest. Once I am clear on that, I would assemble all product stakeholders to generate ideas for product initiatives that align with the OKRs. By initiatives, I mean high-level projects, like a new mobile app, or a new platform feature, like Facebook Live for example.

After assembling a list of initiatives, I would prioritize them based on a cost/benefit analysis. Following the initiative selection process, I would start determining which kinds of features or processes to develop for that initiative. If the initiative is a new platform product, like Facebook Live, then I would put together a list of features to design and develop for that product. I would use insights from UX research, feedback from users, customer support, sales, and any other secondary data source to write user stories. And these user stories will drive the ideation of new features, which I group into themes. Themes help me organize features and identify dependencies.

After compiling a list of features and grouping them into themes, I would prioritize their implementation using a scorecard that evaluates the features based on their contribution of positive impact vs. effort. Attributes that denote positive impact could be, for example, *must have features, frequency of use, ability to leverage existing technology*. And, attributes that denote effort could be *feasibility, engineering complexity* and others. The types of attributes I select for the scorecard depend on user needs and alignment with business objectives (or OKR). Using a Value Point System between 1 and 10, I would assign a number to each feature's attribute. Attributes whose higher values are considered "good" get added, and attributes whose higher values are considered "detrimental" are subtracted. Next, I would calculate a weighted sum of the attribute values for each feature to arrive to a score number. This score number is what I would use to prioritize the features.
Let me summarize this prioritization process with a flow chart:



Flow chart of the prioritization process

INTERVIEWER: Okay. Now imagine that you are asked to prioritize between Stories, Live, and Marketplace as new features for the Facebook mobile app? How would you use your process to do this?
INTERVIEWEE: I would start by asking what the Objectives & Key Results are?
INTERVIEWER: Let's say Facebook wants to increase ad revenue by 10% this year.
INTERVIEWEE: Okay. Stories, Facebook Live, and Marketplace are not small features. They are products. So, I consider them initiatives. Do you agree?
INTERVIEWER: Yes.

INTERVIEWEE: Okay. First I would determine if these initiatives align with the OKR. And they do. The initiatives aim to increase time spent on the platform, and time spent is directly correlated with ad revenue. Facebook Stories aims for the user to return or spend more time on Facebook by viewing or creating stories. Facebook Live wants users to spend time watching real-time videos, such as live sports. And, Marketplace aims for users to spend time on Facebook searching for items they would otherwise find using Google, Craigslist or another tool.

Now that we have established that these three initiatives align with Facebook's OKR, the next step is to prioritize the initiatives based on benefit over cost. For all three initiatives, we can use ad revenue as a proxy to measure benefit. To measure cost, we can use engineering wages as a proxy.

To quantify benefits, I will formulate some equations to estimate ad revenue and engineering labor for each initiative. Then I will compare the initiatives based on benefit/cost ratio, which is ROI, to prioritize them. Does this sound reasonable?

INTERVIEWER: Sure. Go ahead.

INTERVIEWEE: Okay, let's start by talking about the different types of ads Facebook offers and how we can use them to estimate ad revenue. Facebook has various types of ads, but the main ads are: impression ads, click ads, app install ads, page link ads, and video ads. Prices for these ads fluctuate depending on various factors, like bid value, and quality of the ad. So, I am going to use an average price for ad types to simplify my estimates. Do you agree?

INTERVIEWER: Sure.

INTERVIEWEE: Ok. Based on my knowledge of price fluctuations for Facebook ads, the average price of an ad impression is about $0.001 or $1 CPM; the average price of a click or CPC ad is about $0.2, and the average price of a video ad is about $0.01. The price of other ads, like app install and page link, are close to the click ad's price. To simplify my analysis, I will group these ads under the same click-ad bucket. The three types of ads I need to consider are: impressions, click ads and video ads.

Now, I would like to introduce equations that will help me calculate ad revenue for each of the three initiatives.

Let's start with Facebook Stories.

How does Stories generate additional ad revenue for Facebook? When a person gets notified that a friend posted a new story, this may prompt the user to open or return to the Facebook application. So, Stories may contribute incremental ad revenue by displaying impressions or clicks ads when the user opens the app. Here are two equations to estimate impressions and click ad revenue for Stories.

(1) Stories impression ad revenue

Impression Ad Revenue
= (DAU x (%Stories users)
x (%creators)
x (#stories / week)
x (%stories that cause user to open Facebook app)
x (#impressions/story)
x ($CPM / 1000)
x (#weeks / year)

(2) Stories click ad revenue

Click Ad Revenue
= (DAU)
x (%Stories users)
x (%creators)
x (#stories / week)
x (%stories that cause user to open Facebook app)
x (#click ads / story)
x (CTR)
x ($CPC)
x (#weeks / year)

Here's an explanation for these equations:

- There is a number for Facebook's mobile daily active users (DAU).
- A percentage of these users use Stories (%Stories users).
- Of these Stories users, a percentage creates stories (%creators).
- These creators produce some number of stories per week (#stories/week).
- And, only a percentage of these stories cause other users to reopen the Facebook app (%stories that cause a user to open the Facebook app).
- A number of impressions are displayed when the users re-opens the Facebook app to view a story (#impressions/story).
- Each impression generates CPM/1000 of dollars.
- And, since we are calculating revenue for a year, we multiply by the number of weeks in a year.
- To calculate revenue from click-ads due to Stories, we can use the same equation except we replace the number of impressions with the number of click ads. And, replace CPM/1000 with CTR x CPC.

Let's move on to the equation for Facebook Live.

(3) Live ad revenue

Facebook Live generates ad revenue from video ads shown while users watch videos. Here is an equation to estimate ad revenue for Facebook Live:

Video Ad Revenue
= (DAU)
x (%video users)
x (#minutes / day)
x (#video ads / min)
x ($cost of video ad / video ad)
x (#days / year)

Let me explain the equation in more detail.

- A percentage of daily active users (DAU) of Facebook mobile app watch live videos(DAU) x (% video users).
- These live video watchers watch some minutes a day (#minutes/day)
- During these watch minutes viewers are exposed to a number of video ads (#video ads/min).
- And, each video ad has a cost to advertisers ($cost of video ad/video ad).
- We multiply by the number of days in a year to get the total ad revenue per year.

Ok. Let's move on to the equations for Marketplace.

While using Marketplace, a user does not see ads. But, ads are shown to users when they log into facebook and access Marketplace. And, perhaps after using Marketplace, they might

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stay to read their newsfeed. So, ad revenue attributed to Marketplace can come from these instances.
In these situation, I will use these two equations to estimate ad revenue from impressions and click ads.

(4) Marketplace impression ad revenue

Impression Ad Revenue
= (DAU)
x (#Facebook sessions / day)
x (%sessions started due to Marketplace)
x (#impressions / session)
x ($CPM / 1000)
x (#days / year)

(5) Marketplace click ad revenue

Click Ad Revenue
= (DAU)
x (#Facebook sessions / day)
x (%sessions started due to Marketplace)
x (#click ads / session)
x (CTR)
x ($CPC)
x (#days / year)

The details for ad revenue from impressions are:

- Daily active users (DAU) start a number of session per day (DAU) x (#Facebook sessions/day).
- Of these sessions, only a percentage are started by users searching in Marketplace (%sessions started due to Marketplace).
- During each session the user is exposed to a number of impression ads (#impressions/session).
- The cost-per-thousand impressions charged to advertisers is CPM, so the cost for each impression is (CPM/1000).
- To get an estimate for the entire year, we multiply by the number of days in a year.

We can use the same equation to estimate click ad revenue for Marketplace, except we replace (#impressions/session) x ($CPM/1000) with (#click ads/session) x (CTR) x ($CPC).
Now, I will make some assumptions to replace the terms with numerical values.
(The interviewee creates a two column table. The column on the left is to state assumptions and known facts. The right-hand column is to show how to apply these assumptions and facts to the calculations.)
Stories ad revenue

ASSUMPTIONS AND FACTS	CALCULATIONS
	(1) STORIES AD REVENUE FROM IMPRESSIONS
1B Facebook Daily Active Users	= 1B users
10% are Story users	X (10%)
50% create stories	X (50%)
5 stories created a week	X (5 stories per week)
70% of stories trigger users to open app	X (70% stories that cause user to open Facebook app)
10 impressions displayed per story viewed	X (10 impressions/story)
CPM is $1	X ($1/1000 impressions)
52 weeks in a year	X (52 weeks/year)
	= $91M
	(2) STORIES AD REVENUE FROM CLICKS
1B Facebook Daily Active Users	= 1B users
10% are Story users	X (10%)
50% create stories	X (50%)
5 stories created a week	X (5 stories per week)
70% of stories trigger users to open app	X (70% stories that cause user to open Facebook app)
2 click ads displayed per story viewed	X (2 click ads/story)
% ads that are clicked (CTR) is 1%	X (1%)
Cost per click (CPC) is $1	X ($0.2/click ad)
52 weeks in a year	X (52 weeks/year)
	= $36M
TOTAL AD REVENUE FROM STORY	**$127M**

Facebook Live ad revenue

ASSUMPTIONS AND FACTS	CALCULATIONS
	(3) FACEBOOK LIVE AD REVENUE
1B Facebook Daily Active Users	= 1B users
50% watch Facebook Live	X (50% video users)
They watch 5 minutes a day	X (5 minutes/day)
2 video ads are shown every 5 minutes	X (2 video ads/5 minutes)
Cost of a video ad is $0.01	X ($0.01/video ad)
360 days in a year	X (360 days/year)
	= $3.6B
TOTAL AD REVENUE FROM LIVE	**$3.6B**

Marketplace ad revenue

ASSUMPTIONS AND FACTS	CALCULATIONS
	(4) MARKETPLACE AD REVENUE FROM IMPRESSIONS
1B Facebook Daily Active Users	= 1B users
A user opens Facebook 30 times (session) a day	X (30 sessions/day)
0.1% of users use Marketplace	X (0.1% users used Marketplace daily)
10 impressions are displayed per session	X (10 impressions/session)
CPM is $1	X ($1 CMP/1000 impressions)
360 days per year	X (360 days/year)
	= $108M
	(5) MARKETPLACE AD REVENUE FROM CLICKS
1B Facebook Daily Active Users	= 1B users
A user opens Facebook 30 times (session) a day	X (30 sessions/day)
0.1% of users use Marketplace	X (0.1% users used Marketplace daily)
5 clicks per ads are displayed per session	X (5 click ads/session)
% ads that are clicked (CTR) is 1%	X (1% CTR)
Cost per click (CPC) is $1	X ($1 CPC)
360 days per year	X (360 days/year)
	= $108M
TOTAL AD REVENUE FROM MARKETPLACE	**$216M**

So, it looks like Stories would bring in $127M, Facebook Live $3.6B, and Marketplace $216M.

Now, let's look at the cost of developing these platform features to complete our cost benefit analysis.

I will use the cost of engineering labor as a proxy. Of the three platform features, I think Marketplace is the easiest to develop and maintain. Marketplace requires uploading and storing static photos, presenting product information, and enabling users to message sellers about the products they are interested in. There is no video streaming involved, nor functions to transact purchases. So, I would guess that it takes a team of five engineers to build and maintain. Assuming that each engineer's salary is about $200K a year, that means Marketplace costs $1M a year.

Between Facebook Live and Stories, I think Stories is less complexed to develop and maintain. It involves uploading videos, sharing, annotation, and notifications but no live streaming like Facebook Live. Stories are viewed asynchronously. So, I am going to guess that it takes a team of 10 engineers to build and maintain which is 10 x $200K for a total cost of $2M a year.

The real-time nature of Facebook Live makes the engineering requirements more challenging. Facebook Live needs broad and fast network bandwidth, optimized frame buffering, and be synchronous with various functions like audio and feed conversations. So, I am going to estimate that it takes a team of 30 engineers to maintain Facebook Live which is 30 X $200K for a total cost of $6M a year.

Now, let's compare these three initiatives with their benefits and costs with an ROI metric or benefits/cost ratio.

FACEBOOK PLATFORM FEATURE	BENEFIT	COST	ROI (BENEFIT/COST)
Stories	$127M	$127M	127
Facebook Live	$3.6B	$6M	600
Marketplace	$216M	$3M	72

Facebook Live is the platform feature with the highest ROI and would be the one I would select to build. But, we also need to check if this option is likely to meet the mandated OKR. The OKR was to increase ad revenue by 10% YoY. Given that Facebook mobile revenue is expected to be about $30B in 2017, then $3B (10% of $30B) would be the desired incremental revenue. We estimated $3.6B for Facebook Live ad revenue which is $600M more than the expected 10% YoY incremental desired ad revenue. This result favors Facebook Live.

The next step is to start working on the product features for the Facebook Live offering. As I mentioned before, this entails thinking about user needs, creating user stories, and prioritizing them. Would you like me to demonstrate how I would do this for the Facebook Live offering?

INTERVIEWER: No, we are running out of time. However, could you summarize your analysis to wrap up?

INTERVIEWEE: In summary, I explained my process for prioritizing features. It starts with OKRs, and is followed by listing initiatives that are aligned with the OKRs, prioritizing these initiatives based on benefit vs. cost or ROI, listing features for the winning initiative, grouping features into themes, and finally prioritizing features using a scorecard system. I applied this process to the prioritization of three Facebook platform features: Stories, Facebook Live and Marketplace. In this example, these platform features are product initiatives, so I prioritized them as such. In this example, Facebook Live was the winning option with an ROI that was multiple times better than Stories and Marketplace. Therefore, my recommendation would be to build Facebook Live first.

What new feature would you design for Google Daydream VR?

What is this question about?
This product design interview question tests whether you have a sound process for making decisions on which product feature to build next. Your process should include the state of the existing product, the competitive landscape, the users and their needs and how you will use this information to decide which features to build.
What is the interviewer looking for?
The interviewer is evaluating you on the following:

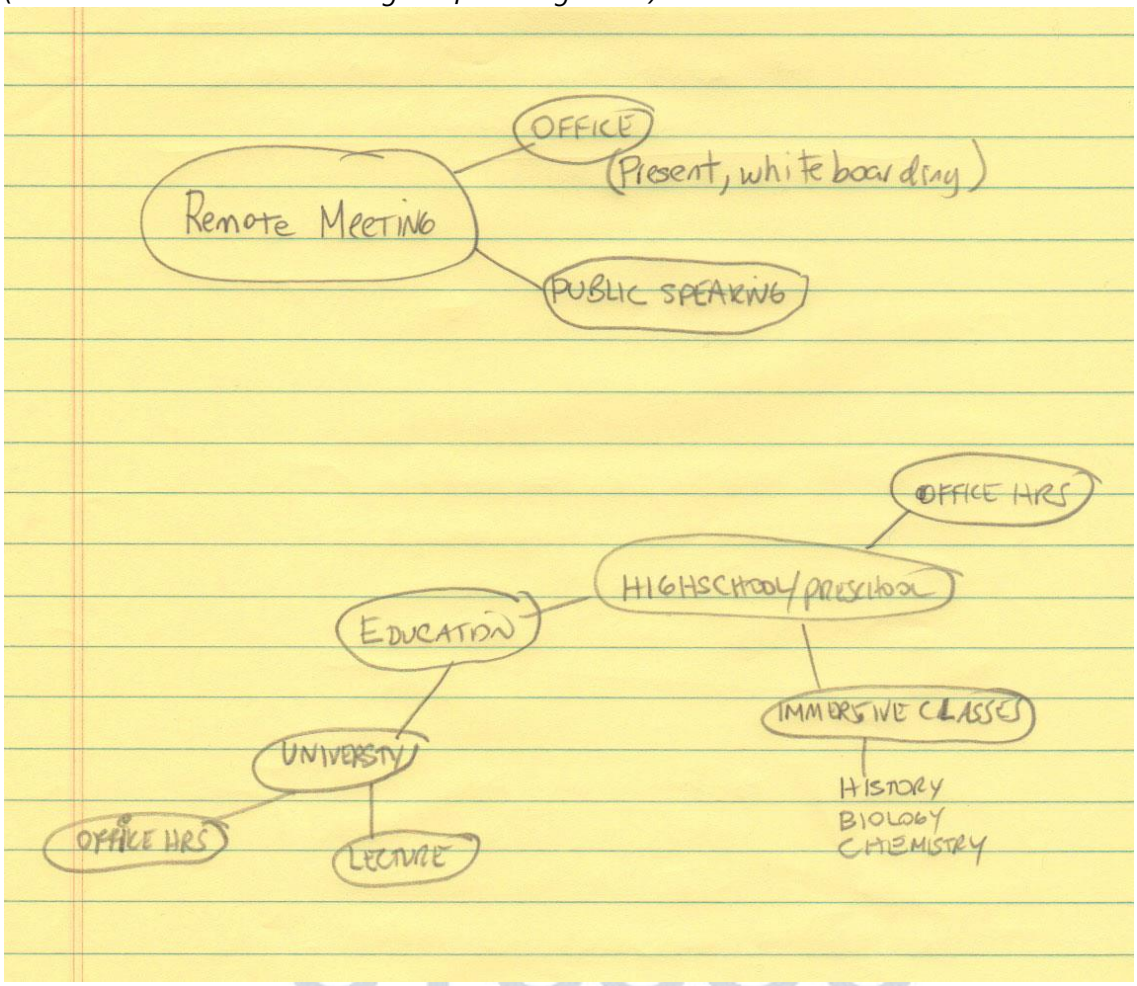
- Is your answer structured and logical, or do you go off on a tangent?
- Do you provide keen insights about the customer and their needs?
- Are you able to provide multiple and diverse use case scenarios?
- Do you go beyond generalities in your solutions and provide detailed descriptions?
- Can you provide ideas that no other candidate has mentioned?
- Are you confident and sound credible?
- Would engineers and product people follow your lead?

How to structure your answer?
We suggest structuring your answer in the following way:

- Describe the product and what it currently does.
- Compare the product with competitive offerings.
- Based on your previous analysis, state what the goal of the new feature should be.
- Brainstorm use cases and prioritize them.
- Brainstorm solutions and discuss tradeoffs.
- Make a recommendation.

Answer Example
INTERVIEWEE: Okay, first I would like to take a step back and talk about the current VR product offerings from Google, and then examine how they stand relative to its competitors. This information will set the stage for a discussion on what the goal for a new future should be. After defining a goal, I will discuss possible use cases to consider, prioritize them, brainstorm on solutions, and make a final recommendation. How does this sound?
INTERVIEWER: Sounds great, go ahead.
INTERVIEWEE: Okay. Google today has two VR products, the smartphone based and the PC-based VR headset, which has a companion controller. Google has announced a future standalone VR headset and future integration of AR to the DayDream platform. Pilots in retail and automotive indicate that Google wants to introduce VR solutions in these markets. For example, Google has partnered with Gap to offer a VR solution to enable people to try on clothes virtually. Google has also partnered with BMW to create virtual showrooms. Being the first to enter these enterprise markets with VR offerings is a good idea. But, I think the consumer market is a bigger opportunity, and the question remains, why would consumers prefer a Google DayDream VR solution over a competitor's solution. Google needs to differentiate itself by solving a key problem only Google has the technologies and wherewithal to solve. Google could leverage its proprietary technologies and leadership advantage in some markets to develop compelling VR applications that leverage these strengths. This should be the objective of the next VR feature.
Use cases
There are two markets in which Google has technological and market share advantages: office applications and education tools. Google office applications are ubiquitous in many

work environments and have been successful at competing with Microsoft. In the education market, Google has surpassed Apple as the number one seller of devices. And not only devices but software too. Google has penetrated this market substantially in the US with its G Suite for schools.
Because of these advantages, I think these two markets are worth exploring for possible VR applications. Now I would like to brainstorm possible use case scenarios. Could I have a minute to explore some ideas?
INTERVIEWER: Sure.
INTERVIEWEE:
(The interviewee starts drawing the following chart.)



Brainstorming use case scenarios.
Ok, I have some possible use case scenarios. Ones that come to mind involve offices and public schools.
Offices
Here are some use case for the office scenarios:

- **Remote meetings**—employees working in different locations often have the need to meet as a group. Web conferencing solutions exist today but VR could improve the experience by placing people in the same room to collaborate more easily and make meetings more enjoyable.
- **Public speaking**—VR solutions could enhance the experience by anonymizing people's identities and thus making it easier for people that are shy to participate.

- **Front desk**—instead of dealing with customers over the phone, a VR environment could make the experience more enjoyable. For example, talking face to face with an avatar as opposed to voice without a face would improve customer service because the caller gets visual feedback of the intake process.
- **Waiting on hold**—most callers to customer service dislike having to wait on hold for extended periods of time. Instead of playing music while on hold, create a VR experience for the caller. For example, a business could create an enjoyable branded experience, which could decrease perceived waiting time.

Public schools

Some use case scenarios for public schools are:

- **Teaching classes**—most classroom settings are passive. A student sits and listens or watches a lecture. Instead, VR could provide an active classroom setting by providing an immersive learning experience. I can see classes like History, Biology, or Chemistry becoming more engaging and memorable with VR.
- **Office hours**—conducting office hours in a virtual environment could have many benefits. For example, virtual office hours would be more convenient by saving travel time to students. It might also be an easier environment for shy students to ask questions.

These two markets have different needs, so I will choose the one with the best opportunity. One criterion is market size based on the number of possible users. Let's do a quick estimate of the potential number of users in the **office applications market** and **public education market**. I will estimate this for the US only, since the US is likely to be the first market to adopt VR.

Let's start with office applications users. From my general knowledge of the labor market, I know that the number of people working in the US is about 130M. Of those, I will assume that 50% are office workers, which results in 65M users.

To estimate public school users, I know there are about 100,000 public schools in the US. Since schools come in different sizes, I will segment them into large, medium and small to estimate more accurately. I will assume 30% for large, 20% medium and 50% for small. To calculate the total number of students, I will assume:

Students in Large Schools
= [% of large schools]
X [# of public schools in the US]
X [# of students/large school]
= 30% * 100,000 * 1,000
= 30M students in large schools

Students in Medium Schools
= [% of medium schools]
X [# of public schools in the US]
X [# of students/medium school]
= 20% * 100,000 * 500
= 10M students in medium schools

Students in Small Schools
= [% of small schools]
X [# of public schools in the US]
X [# of students/small school]

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= 50% * 100,000 * 200
= 10M students in small schools
Total Students

- 30M students in large schools
- 10M students in medium schools
- 10M students in small schools
  = 50M

There are 15M more office applications users than public school users. So, I am going to focus on VR solutions for office environments, since it is a larger market.
Does this sound reasonable?
INTERVIEWER: Yes, please proceed with ideas for a new feature.
INTERVIEWEE: Sure.
Okay, now I would like to go deeper into the use cases for office environments. The three scenarios I suggested were: remote meetings, public speaking, and front desk. I think having remote meetings is a frequent use case in offices and one for which a solution would have a big impact on employees. So, I would pick this use case as the one to focus on and go deeper into tasks that people need to perform. This will help think of new features to support these tasks.
In meetings, I think the following tasks are common:

- **Presenting slides**—The need to show slides during a meeting.
- **Share your screen**—The need to make a laptop screen visible to others in real time; or the need to make a document visible to others in real time to work collaboratively.
- **Whiteboarding**—The need to write or sketch to brainstorm ideas.
- **Taking notes**—The need to take notes during a meeting.
- **Email and Messaging**—The need to read email or send a message.
- **Surfing the net**—The need to perform online searches.

Some ideas for features to build to help accomplish these tasks:

- **Presenting slides.** Simulate a room with a table, chairs and a whiteboard to project the presentation. Place the chairs in front of the whiteboard so attendees can see the presentation without obstruction. A side menu should present options to select the presenter and to zoom in or out of the presentation since people have different eyesight levels. Use an external control of the VR headset to select the menu options.
- **Sharing screens.** In meetings today, people can share their screens with others via web conferencing applications. I can see people wanting to replicate this capability in a VR meeting. One idea is to place simulated laptops in front of the avatars and enable them to share their screens with others. Another option is to allow avatars to project their screen to the whiteboard. Use a virtual mouse and keyboard to control an avatar's virtual laptop.
- **Whiteboarding.** Have avatars use simulated pens to write or draw on a simulated whiteboard in the scene. Multiple colors and stroke widths should be available as menu options. And, the whiteboard should provide a choice to save content to an avatar's Google Drive account.
- **Taking notes.** An avatar could take notes by hand or by typing. Enable handwriting by providing an external surface where the person could write and have their handwriting display inside the VR environment. The notes could be stored on the avatar's Google Drive account.

- **Email and messaging.** These could be communication features that are available on all VR applications, not only office applications. Email and messaging should be permanent options of a systems menu. An external keyboard could be used to facilitate typing. The email and messaging screens should only be visible to the avatar doing these tasks, to maintain privacy.
- **Surfing the net.** Searching the web while in meetings is very common. If people are in a VR meeting, taking the headset off to browse in the computer would be inconvenient and disruptive for the user. So one solution is to have a menu option to surf the net. Selecting this option would open a Google Chrome browser, putting the meeting scene in the background. An external mouse and keyboard can be synchronized with the VR scene to enable the user to navigate the web easily.

I would like to prioritize some features since there are too many to build at once. Presenting slides and whiteboarding are at the top of my list since they occur the most during meetings. The whiteboarding feature would probably take less time to build because the ability to write or draw with a virtual pen within the VR environment is already feasible. So, I would start with the whiteboarding feature first. Build it, launch a pilot, and get feedback from users about the experience. These learnings will be useful for a future implementation of the 'slide presentation' feature.

Google could make the whiteboarding and the slide presentation VR features part of its Google G Suite, perhaps as additional applications. Google could be the first to offer VR office applications. Microsoft could compete in this space, but I think they would have some big hurdles to overcome. First, OneDrive is third in market share behind Google Drive; second, Google dominates the public schools market where future generations of office workers are already using Google office applications; and third, Microsoft is behind Google and Facebook in the VR market.

In summary, I think Google should leverage its proprietary technologies and market leadership in the office applications space to provide VR solutions that no other competitor can match. The office applications market is large and growing; it is a huge opportunity.

I proposed and prioritized use case scenarios in the office applications market, opting to focus on the remote meeting scenario. In this scenario, I listed several tasks that could be enhanced with a VR solution, among them, making a slide presentation and whiteboarding. I recommend starting with the whiteboarding solution because it would be faster to develop.

Although Microsoft is currently number one in the office applications space, Google has advantages that could make it number one in VR office applications. Google has stronger leadership in the cloud storage market, large penetration with young people starting to learn how to use office applications, and market leadership in VR over Microsoft.

How would you improve Facebook's search monetization?

What is this question about?
This product execution interview question tests whether you can decompose a problem, think in a structured way and articulate your solutions.
How to structure your answer?

- Describe how Facebook currently does search and how it monetizes it.
- Analyze how Facebook compares to Google and mention relative strengths and weaknesses.
- Describe use case scenarios that can take advantage of Facebook's strengths.
- Prioritize use case scenarios to implement.
- Ideate solutions for these scenarios.
- Think of ways to monetize these solutions.
- Wrap up

Answer Example
INTERVIEWEE: Facebook currently has some search capabilities, and my impression is that they are monetizing it through search ads. Is this correct?
INTERVIEWER: Yes.
INTERVIEWEE: So, we are looking for ways to increase revenue generated through search ads?
INTERVIEWER: Well, search ads are one way to generate revenue. But, I would like to hear from you other ideas beyond search ads.
INTERVIEWEE: Okay, understood. So I would like to start by doing an overall analysis of search in Facebook today vs. the competition, then describe user search scenarios, prioritize them, and brainstorm ideas on how to monetize them in ways beyond search ads.
INTERVIEWEE: Since I have not done many searches in Facebook on the past, I would like to run a few now to understand better how it works. Could I take a few minutes and do that now?
INTERVIEWER: Of course, please go ahead.
INTERVIEWEE:
(The interviewee runs two searches, one to search for 'running shoes for women' and the other to search for 'auto maintenance services.')
Competitive analysis
Okay, I just did two sample searches: one for finding women's running shoes and another for auto maintenance services, which I currently need. From what I observed, the results are not actionable. The search organizes the results into three sections: Marketplace, Photos, and Links. The Marketplace results are for used shoes, so it is not relevant since I want to buy new shoes. The Photos section lists posts from people with photos of shoes, but that is not what I am looking for. These results are not useful for several reasons:

1. the search engine is not taking **context** into account, for example, it should deduce that I am looking for new shoes, not used ones, and provide useful information like price and recommendations;
2. the results are **not exhaustive**; the user cannot rely on the results to make an informed decision,
3. the results are **not personalized**, at a minimum, the results should take into account my location,

4. and there is **no prioritization** given to the results, for example, listing by lowest price and the highest number of recommendations is something users expect today.

As a comparison, if you run the shoe query in Google the results are shown within a photo carousel with brand, store, price and recommendations information. You see multiple options to make comparisons that helps you narrow down which shoes to buy. In the case of searching for auto maintenance services, Google displays the closest auto repair shops around me. These results are useful. They enable the user to narrow options to choose from, a very actionable result.

Hurdles

Now, before talking about search use case scenarios, I need to point out that unless the search algorithm is improved, it is unlikely users will see Facebook as a search destination. The Facebook search engine needs to prove its value to users first.

Moreover, another hurdle is the fact that Google is the de-facto top-of-mind destination for search. Changing people's mindset about Facebook as a search destination would require changing people's habit of always using Google.

I think Facebook should leverage its advantage in knowing personal and granular behavioral data about its users to provide better search results. For example, users favor recommendations from friends when searching for products, services or entertainment. And they like detail experience descriptions about a product or service, something Facebook users already shared with posts. Google could try to do the same using Google+, but as we know Google+ has not taken off.

With this in mind, I would like to brainstorm about different user search scenarios and prioritize the ones that could leverage Facebook's social network data.

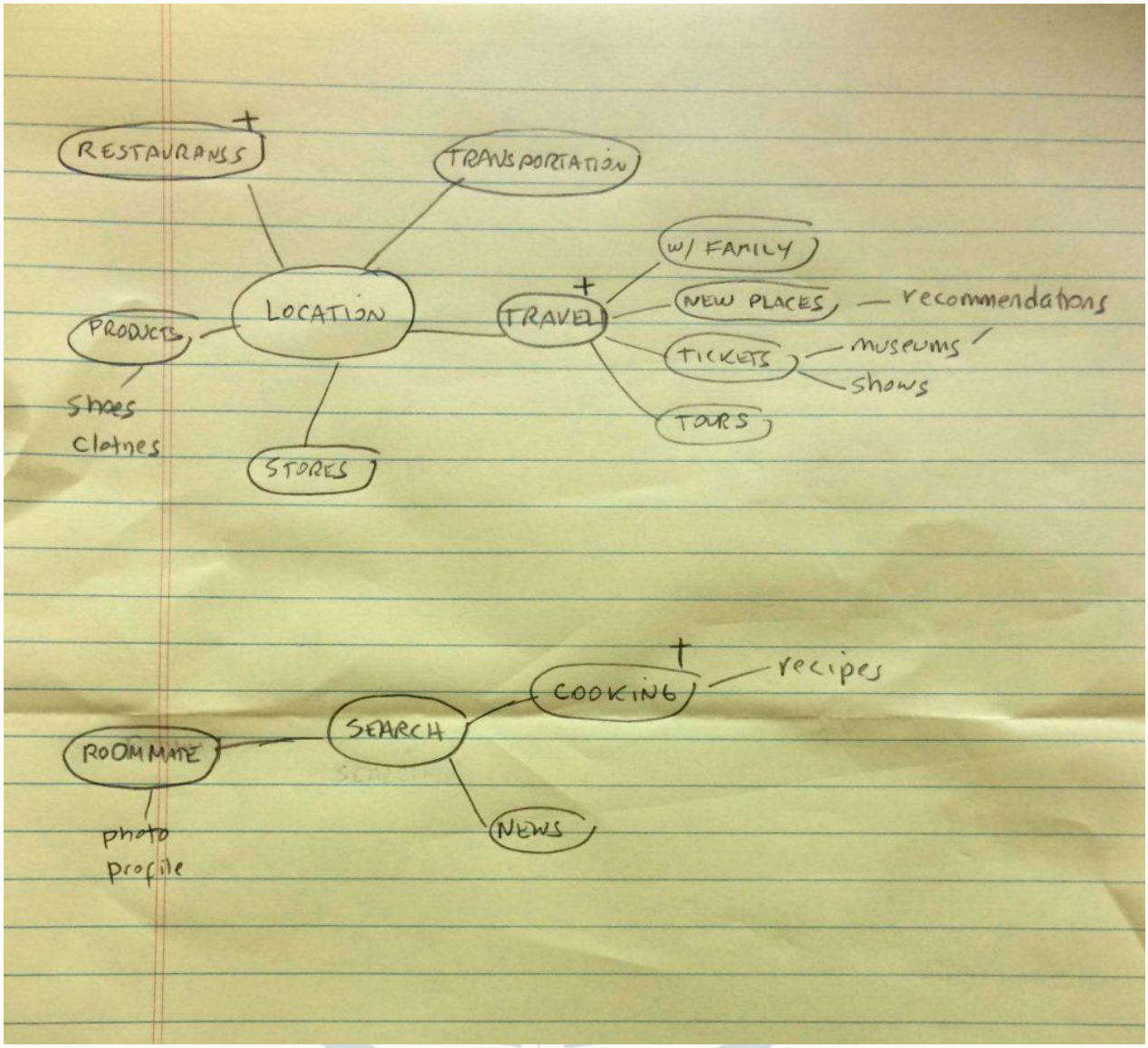
Could I take a minute to think about possible use case scenarios?

INTERVIEWER: Sure

INTERVIEWEE:

(The interviewee draws the following word association graph to think of different use case scenarios. She marks with + those that will be more fruitful.)

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Word Association Brainstorm

Okay, here are a few use cases that I think would be fruitful to consider.

- People traveling to a new city or country for tourism usually need help in finding places, events or restaurants in the area. Recommendations from friends or a large pool of people, with photos, and detail entries of their experience, would be useful.
- A second scenario is helping users find good cooking recipes recommended by the user's friends or other Facebook users.
- A third scenario is looking for a good personal service provider, like a stylist or massage therapist. Having recommendations from friends or others about the person that directly provides these services, rather than recommendations on the business they work, are better.
- A fourth scenario is finding possible roommates profiles that meet a user's criteria.

I would prioritize these scenarios based on which ones leverage Facebook's proprietary social data. Of these four, I think recommendations while traveling and finding a personal service provider meet these criteria. But, I prefer the traveling scenario, since it is more suited for a mobile search, and mobile access to Facebook is greater than 50%.

Solutions

In the traveling scenario, Facebook could provide search results to restaurants, events, and places to visit based on friends' recommendations. Perhaps recommendations from previous visits to the same city or maybe recommendations of what to do in that city. The Facebook search engine would need to know which friends had visited the same place and whether they posted about them. If there are not too many postss from friends, the search algorithm could aggregate data from all Facebook users that visited the same location. One concrete implementation for the mobile app could be to add a new entry under the Explore section and call it Travel Companion. After tapping the Travel Companion option, a map appears with pins close to the user's current location, highlighting places such as restaurants, museums, bars, hotels, and theaters. The map could also show photo bubbles of friends that have been at different locations in the same city. A filter control could be added to let the user choose which types of places to show on the map. The map could be displayed in two modes: full screen or half screen, with the lower half presenting more details about the pinned locations. Tapping on a theater or museum list entry could take the user to a page that shows the current show times and interface to order tickets. A similar idea can be used for restaurant reservations.
Now, regarding how to monetize this feature, I think a transaction, search ad, and call-to-action model can be used.
Here are some examples:

- When a user buys a ticket from the theater or museum, a transaction fee can be charged from those establishments.
- Charge a search ad fee if the user clicks on a sponsored ad that appears as part of the search results.
- Charge a call-to-action click fee when the user clicks on a CTA button in a search result.

For this type of search feature to be successful, there needs to be a lot of data available from Facebook users. Finding posts with location information is not a problem since users are in the habit of tagging posts with their location. But, getting users to enter recommendations about places they visited could be harder especially once the experience is over. One solution could be to send reminder notifications to users that bought tickets or eat at a restaurant to provide feedback. The feedback can take the form of a simple simple star rating system with additional space for comments.
Would you like me to elaborate on this?
INTERVIEWER: I think we are running out of time so how about just summarizing your analysis.
INTERVIEWEE: Sure. Today, Facebook is not top-of-mind as a place for doing searches, Google is. The search results that the Facebook search engine provides are not very useful. To change users' mindset about Facebook as a search destination, Facebook needs to improve its search engine. This is a precondition to monetizing search in Facebook. One way of improving search is by leveraging Facebook's knowledge of personal and detail behavioral data about their users. The types of search queries that could benefit from this knowledge are recommendation-based searches. I brainstormed different scenarios and proposed a solution for a typical scenario: a traveler going to a new place and looking for recommendations for places to visit, restaurants, etc. The solution consists of displaying a map with photo bubbles of friends and pins of locations they visited close to where the user is. If the number of friends is too small, the results can be complemented with other Facebook users' recommendations. Finally, I suggested three monetization models that can

be applied to this feature: transaction, search ad and call-to-action fees. I think this solution can be used as a proof of concept that Facebook can provide better search results than Google, when social data is key to finding targeted results.

What new feature would you design for the Amazon Echo?

What is this question about?
This product design interview question tests whether you understand the process of going from customer needs to product development. This process involves determining who the customer is, knowing what they want to accomplish, defining multiple use case scenarios, and prioritizing what to build.

What is the interviewer looking for?
The interviewer is evaluating you on the following:

- Do you provide keen insights about the customer and their needs?
- Are you able to provide multiple and diverse use case scenarios?
- Is your answer structured and logical, or do you go off on a tangent?
- Do you go beyond generalities in your solutions and provide detailed descriptions?
- Can you provide ideas that no other candidate has mentioned?
- Are you confident and sound credible? Would engineers and product people follow your lead?

How to structure your answer?
One way to structure your answer is by using the CIRCLES Method™ by Lewis C. Lin. A detailed description of this method is in the book *Decode and Conquer*. This approach is useful in helping you think and answer in an organized and exhaustive way. The steps of this method are:
C—**Clarify**. Ask clarifying questions to narrow the scope.
I—**Identify** the users/customers as personas such as food lovers, soccer moms, etc.
R—**Report** on the user's needs (use cases). A use case is an activity that a user would like to do relative to the product.
C—**Cut** through and prioritize the use cases based on attributes such as revenue, customer benefit, complexity, etc.
L—**List** solutions.
E—**Evaluate** the tradeoffs of your solutions.
S—**Summarize**:

- State which solution you would recommend.
- Recap what the solution does and why it is beneficial.
- Explain why you prefer this solution.

How to quickly think of several use cases?
In a previous article, we described three methods that can help spark ideas for use cases. The methods are the 5Ws, word associations, and SCAMPER. In this example, we will use the Word Associations method.

Answer Example
INTERVIEWEE: What is the goal that the business wants to achieve?
INTERVIEWER: We want to increase sales of the Amazon Echo and the orders of products and services from Amazon.
INTERVIEWEE: Just to be clear, the new features are just for the Amazon Echo and not for other voice-controlled assistant devices like the Amazon Dot, Amazon Show nor Amazon Look?
INTERVIEWER: Yes, the feature is only for the Amazon Echo.
INTERVIEWEE: Okay, I would like to start by talking about the types of users that are likely to buy voice-controlled assistant devices like the Amazon Echo, who they are and what

things they may want to do that the current Amazon Echo product does not support. After brainstorming on some use cases related to voice-controlled assistants, I will select the ones that I think would help achieve the goals if backed by a new feature.

INTERVIEWER: Sounds good.

INTERVIEWEE: The main types of users of the Amazon Echo are:

- Families that are technology savvy and have mid to high disposable incomes to buy non-essential electronic devices. Most of these families are also likely to be Amazon Prime members lured by some of the free services for Prime members that come with the Echo. For example, Prime Music is free for members.
- Elderly parents of potential buyers of these devices are potential users. People with elderly parents may buy these devices to aid elderly parents with some tasks.
- Single individuals that are also tech savvy, with high disposable incomes, and are intrigued by the novelty of these devices.

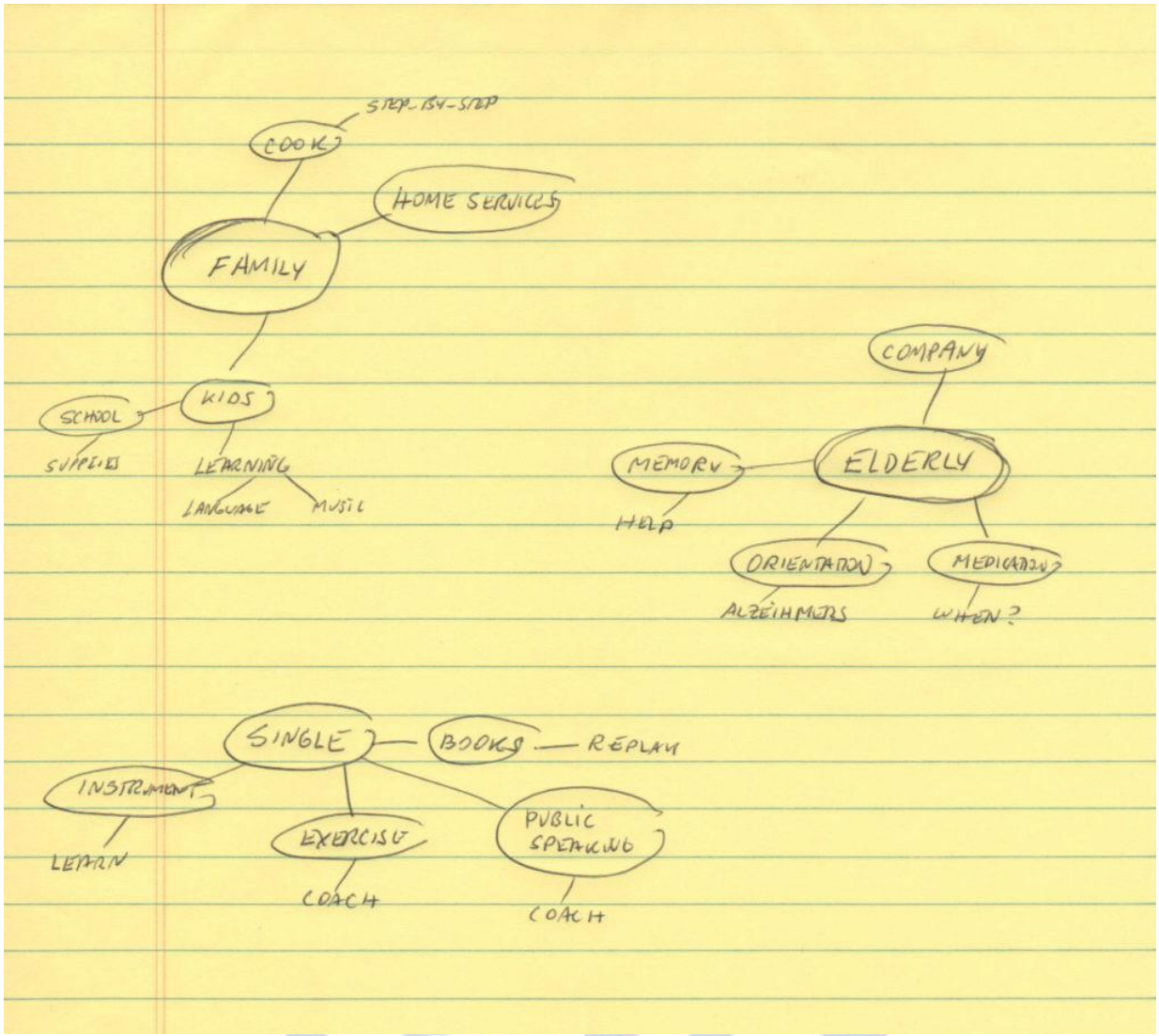
Families and single individuals are buyers and users of these devices while elderly parents are most likely just users.

Okay, I would like to take a minute to brainstorm possible use cases that have not been addressed by Amazon Echo for these three users yet.

INTERVIEWER: Sure.

(The interviewee takes his notepad and starts drawing word associations related to home automation.)

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Brainstorming notes

INTERVIEWEE: Okay, let's start with families.

Families:

1. Families today are too busy and do not have time to plan their meals, such as deciding what to cook, which ingredients to buy, and how to cook the meal. Assistance in choosing a recipe, buying ingredients, and step-by-step instructions on how to cook would save them time. A recurring scenario to address is opening the fridge and wanting to use up miscellaneous food items, but not knowing what kind of meal to make with them. (Association: Family—Cook).
2. Getting help finding a service person to help with repairs or house chores without having to spend much time looking for a trustworthy person. (Association: Family—Home Services).
3. Parents whose children are learning a second language and want a language coach for their kids to help with pronunciation practice. (Association: Family—Kids—Learning—Language).

Elderly:

1. Elderly people who take more than one medication may need help remembering which medication to take, at what times, and in which order to take them. (Association: Elderly).

2. Families with members that have dementia, such as Alzheimer's. Moments of short-term memory loss can cause disorientation and high stress. People with dementia may have trouble remembering where a family member is or when they will be home. During these moments, a typical situation could look like this: a daughter is grocery shopping, while the daughter is away, her mother with Alzheimer's has forgotten that her daughter went to the store. By having an assistant to remind the mother where the daughter is could help alleviate stress.

Single individual:

1. Public speaking coaching is something many professionals are interested. (Association: Single—Training—Public Speaking).
2. Having a personal trainer for exercising.
3. Learning how to play an instrument. Playing by ear requires listening to a passage in a song and trying to replicate it with your instrument again and again. Using your voice to command different passages to practice would be useful so that you can keep your hands on the instrument.

Of all these use cases, I would prioritize the ones that are more likely to help achieve the goals stated earlier: to increase purchases of the Echo and increase purchases of products and services from other Amazon businesses. The use cases that would be aligned with these goals would be those that are important to the user and frequent enough for the user to want to buy the Amazon Echo. Ideally, they should also lead to the frequent purchase of products and services from Amazon. Therefore my criteria for prioritizing use cases is whether it is an important use case, it is a frequent one and whether it leads to purchases from Amazon.

Of these eight use cases, the **cooking instructions** for families, the **medication reminder**, and the **memory aid** for elderly Alzheimer patients are scenarios that frequently occur in the lives of those users. Now from the importance standpoint, I think assistance for elderly parents is of higher importance than cooking. Nevertheless, cooking is an activity that can be more conducive to purchases from Amazon, because of required ingredients and cooking paraphernalia. The elderly use cases could lead to Echo sales, but the cooking activity could lead to more long-term purchases of other products from Amazon. So I would prioritize the cooking use case for development, but put the elderly uses cases in the product roadmap. With aging population increasing, more families are in need of affordable ways to take care of their elderly parents. I think the market for technology that aids the elderly will be growing, and getting families to buy the Echo for their parents is the first step to entering that market.

Solutions

Cooking Instructions

So now, I would like to brainstorm solutions for the cooking instructions case. As I mentioned earlier, one use case is finding what to cook from miscellaneous food items in the fridge. One solution could be to provide a Chef Assistant skill that a user could ask for advice. The user could list the ingredients in the fridge and then ask Alexa, "Alexa, what can I cook with these ingredients?" The skill could provide the top three recommendations, from which the user could select one, and then ask Alexa to order the missing ingredients from Amazon Fresh or Prime Now. When the user is ready to cook, the Chef Assistant skill could provide instructions on prep, such as organizing and chopping ingredients, then give step-by-step instructions on how to cook the actual meal. If the user misses a step, a voice-enabled search and playback controls should be available for the user to repeat the

instructions. For example, a user may say, "Alexa, start from the moment the chicken breast is seasoned." An alternative solution or a companion to this solution is to display a video of the cooking instructions to the browser or mobile Alexa app. The video playback and audio playback from Alexa should be synchronized so that the instructions the user hears corresponds to what the user is seeing.

Memory Aid

Now, I want to brainstorm a solution for the elderly that have short-term memory loss or dementia, such as Alzheimer's. In my use case example, a mother forgets that her daughter went grocery shopping. Picture a daughter saying, "Mom, I going to the store and will be back in one hour." The daughter leaves. Some time goes by, and the mother calls out for her daughter, but there is no response. The mother has forgotten that her daughter is at the store and begins to worry. In this type of situation, the daughter's name could be an invocation name. When the daughter's name is called out, Alexa wakes with the response, "Mary is at the store and will be back by 3:00 PM. Would you like me to call her?" This solution would require the Echo skill to learn who the family members are, their relationships, their phone numbers, and track their locations through their smartphones. It would be useful to have voice identification so that the individual with Alzheimer's doesn't need to use "Alexa" as the wake word. Alexa could identify the person with dementia and would know what to say to this person. An Echo skill could be developed by Amazon or a third party to provide this kind of service.

Medication Reminder

A medication management skill could be developed to remind an elderly person to take their prescriptions. A person could ask Alexa to tell them which pills to take, how many to take, and at what time. Before taking medication, Alexa could verify if the person has taken the prescription. Alexa would log whether they did or not. For example, Alexa could say, "Paul, it is time for you to take your Atenolol. Have you taken it already." If Paul replies with a "Yes" Alexa could respond with "Great! I will record that in your log." If Paul replies with a "No" Alexa would remind him to take his medication. After a few minutes, Alex would follow-up with, "Paul did you take your Atenolol." By reminding and logging medications, Alex could keep track of which meds to take next.

Evaluation

Of the two solution proposals for the cooking use case, I would start with the one without video. It is simpler to make and doesn't require the synchronization between video and audio. Besides, with the Echo Show now in production, it would be better to have the video version of this skill developed for the Echo Show alone, because the implementation would be easier, cleaner and would increase the Echo Show appeal.

Of the two solutions to support the elderly, I would implement the one that assists short-term memory loss or dementia, such as Alzheimer's. Although it would be harder to implement, because it requires the coordination of multiple technologies like real-time location tracking and voice identification, I think the solution would have a much stronger impact. The solution would reduce stress for the person with short-term memory loss and the people that care for them. If implemented correctly, this solution could open the door to addressing more use cases in the care management industry.

Summarize

In summary, I have explored use cases that the Amazon Echo has not yet addressed. Of the eight I proposed, I would recommend implementing the cooking instructions use case, because it is very frequent, important to the user, and is likely to increase purchases from

Amazon. One solution I suggested provides the user with recipes for their miscellaneous food items in the fridge, and step-by-step instructions with voice controlled playback and search capabilities. In the product roadmap, I also recommend the implementation of the memory aid solution for short-term memory loss. Although this solution may not lead to many commercial purchases of Amazon products, it can open the door to a new market in care management, which is a $15B market and growing.

How would you estimate mobile ad revenue for Facebook?

What is this question about?
The interviewer is assessing your ability to provide quick estimates on any business related issue. The question could be about the size of a market, revenue estimates, number of daily active users, etc.
What is the interviewer looking for?
In product estimation questions, interviewers are evaluating your problem-solving and quantitative skills. They are looking for a ballpark number, not an accurate number. What matters is whether you are logical, can explain all your assumptions clearly, are organized in your work, and good with numbers.
Answer Structure

- **Ask clarifying questions.** Clarifying eliminates any ambiguity of what should include in your calculations.
- **Make an equation.** Consider edge cases or alternate sources of data and writing any facts that you know helps with calculations.
- **Break down the equation into components.** Write your assumptions next to the components. Doing this reminds you to explain the assumptions clearly to the interviewer.
- **Do the math.** Calculate the result of each component and compute the result.
- **Do a sanity check.** Do your results make sense? If not, recheck your equation, assumptions, and arithmetic.

Answer Example
INTERVIEWEE: Should I estimate mobile ad revenue for the US or the entire world?
INTERVIEWER: For the world.
INTERVIEWEE: Okay, thanks. I know that Facebook currently offers four bidding options as ways to pay for ads: cost per 1,000 impressions (CPM), cost per link-click (CPC), cost per mobile app install and cost per Page like.
Facebook's ad buying system is an auction-based bidding system. The winning bid depends on different factors besides the highest bid. It also depends on the quality of an ad, the target audience, the target industry, the country, and location where the ad appears among other properties. On mobile, the location where the ad appears is going to be the newsfeed or inside an app.
CPM, CPC, cost per mobile app install and cost per Page like, vary according to these factors. The same factors affect click-through rates (CTR)—the number of clicks divided by the number of impressions per ad.
I will use these metrics to estimate global revenue for mobile ads. But, to make a quick estimate, I need to use single numbers for these metrics; therefore I will use estimated averages in my calculations. Does this sound reasonable?
INTERVIEWER: Yes, go ahead.
INTERVIEWEE: Okay. Revenue from ad impressions and ad clicks can be calculated using two different equations. The impressions equation is based on cost per 1,000 impressions (CPM) while the clicks equation is based on click-through rates CTR and cost per click; where the click can be a link click, a mobile app install click or a Page like click.
Equation A: *Annual revenue from impressions:*
[$ / year] =
[#impressions / 1 person] X

[#persons / day] X
[cost per 1,000 impressions (CPM) / 1000 impressions] X
[360 days / year]

Equation B: *Annual revenue from clicks:*

[$ / year] =
[#clicks / #ads] X
[#ads / day] X
[cost per click] X
[360 days / year]

There are different click-through rates and cost per click for the Link click, App Install and Page like bidding options, so the revenue for clicks equation can be split into three:

Annual revenue from link clicks:

[$ / year] =
[CTR] X
[#persons per day X #link-click ads / person] X
[CPC] X
[360 days / year]

Annual revenue from cost per mobile app install:

[$ / year] =
[app install rate] X
[#persons per day X #app install ads / person] X
[cost per mobile app install] X
[360 days / year]

Annual revenue from cost per Page like:

[$ / year] =
[page like rate] X
[#persons per day X #Page like ads / person] X
[cost per Page like] X
[360 days / year]

Now, I will use these equations to estimate the annual revenue of mobile ads. To count for mobile revenue alone, I will count only the daily mobile users of Facebook. From my recent reading of Facebook news, I recall that the number of daily mobile users is about 1B.

INTERVIEWER: Sounds good.

(The interviewee starts making the calculations below and making side notes about the assumptions he is making as he enters numbers into the equation. Once he finishes each calculation, he walks the interviewer through his calculations.)

ASSUMPTIONS AND FACTS	CALCULATIONS
One person is exposed to 25 viewable impressions per day on their mobile 1B daily mobile users CPM = $1 Year has 360 days	Annual revenue from impressions = [25 impressions per person] X [1B persons / day] X [$1 for thousand impressions / 1000 impressions] X [360 days / year] = ~$9B / year
CTR = 1% 1B daily mobile users A person sees 10 link-click ads CPC = $0.2 Year has 360 days	Annual revenue from CPC = [1%] X [1B persons / day X 10 link-click ads / person] X [$0.2] X [360 days / year] = $7B / year
Mobile app install rate = 0.01% 1B daily mobile users A person sees 2 app install ads Cost per mobile app install = $0.8 Year has 360 days	Annual revenue from cost per mobile app Install = [0.01%] X [1B persons / day X 2 app install ads / person] X [$0.8] X [360 days / year] = $58M / year
Page like rate = 0.07% 1B daily mobile users A person sees 5 Page like ads Cost per Page like = $0.8 Year has 360 days	Annual revenue from cost per Page like = [0.07%] X [1B persons / day X 5 Page like ads / person] X [$0.8] X [360 days / year] = $1B / year

Review and state your assumptions

INTERVIEWEE: Let's start with annual revenue from viewed ad impressions. I check Facebook on average twice in an hour when I am not with my phone or exercising. So, say I am active with the phone 12 hours a day, and I check Facebook twice per hour, that is 24 times. Assuming I am shown an ad impression every time I log into Facebook then that makes 24 times I see an ad impression. Does this sound reasonable to you?

INTERVIEWER: Let's go with it.
INTERVIEWEE: Okay, so 24 ad impressions multiplied by 1B persons results is 24B impressions per day. Now, CPM will vary widely, but I recall reading a report where a global average of $1 was estimated, so I am going to assume $1 for CPM. Multiplying 24B impressions per day by a cost per thousand impressions of $1 and dividing by 1000 results in $24M per day. And, multiplying that by 360 days results in $8,640M or about $9B a year.
Annual revenue from impressions
= [24 impressions per person]
X [1B persons / day]
X [$1 for thousand impressions / 1000]
X [360 days / year]
=~ $9B / year
Let's estimate annual revenue from link-click ads. Again from my readings about Facebook ad benchmarks, I will assume a CTR of 1% and CPC of $0.2. Now, to estimate the number of link-click ads Facebook shows a person per day, I am going to guess that it is half the number of viewed ad impressions for which we just calculated revenue. Half of 24 is 12, so I am going to guess 10 link-click ads, to simplify things.
Applying these numbers to the equation we have 1% CTR X 1B persons/day x 10 ads/person X $0.2 CPC X 360 days = 10^7 X 2 X 360 = 720 x 10^7 or about $7B a year.
Annual revenue from CPC
= [1% clicks / link-click ad]
X [1B persons / day X 10 link-click ads / person]
X [$0.2 / click]
X [360 days / year]
=~ $7B / year
Let's estimate annual revenue from cost per mobile app install ads. In general, I think users click less on ads to install apps than other ads, so I am going to assume a very low CTR of 0.01%. Now, I know that cost per click for app install ads are higher than regular link-click ads, so I am going to assume a cost of $0.8. I am guessing the reason why cost per app install is so much higher than a regular link-click ad CPC, is because the click-through rate of app installs is so much lower than link-click ads. The lower the click-through rate, the less value the ad has for a user since they are not clicking on them that much. Facebook's bidding algorithm takes this into account because it tries to prevent unappealing ads from appearing on Facebook. Now, how many times are these ads shown to a user? I am going to guess twice a day. When I use Facebook, I rarely see these types of ads.
So putting these numbers together to calculate annual revenue from cost per mobile app install, we have 1% CTR X 1B persons/day x 2 app install ads X $0.8 cost per mobile app install ad click X 360 days = 10^5 x 2 x 8 x 36 = 10^5 x 576 which is about $58M a year.
Annual revenue from cost per mobile app Install
= [0.01% clicks / app install ad]
X [1B persons /day X 2 app install ads / person]
X [$0.8 / click]
X [360 days / year]
=~ $58M / year
Let's estimate annual revenue from Page Like ads. I think the click-through rate for Page like ads is higher than app-install ads but much lower than regular link-click ads, so I am going to guess a CTR of 0.07%. In terms of cost per page like, I know the cost is also more

expensive than regular link click ads, so I will use the same $0.8 estimate as the app-install ads. And in terms of how frequently users are shown these Page Like ads, I think they are shown more frequently than app-install ads, but less than link-click ads, so I am going to guess 5 times per person per day.

So putting these numbers into the equation, we have 7% CTR X 1B persons/day X 5 ads X $0.8 cost per Page like ad click X 360 = 7 x 10^5 x 4 x 360 which is about 10^5 x 7 x 1600 or 11200 x 10^5 = $1B per year.

Annual revenue from cost per Page like

= [0.07% clicks / Page like ad]
X [1B persons X 5 Page like ads / person per day]
X [$0.8 / click]
X [360 days / year]
=~ $1B / year

So summing all these numbers up, we have $9B + $7B + $58M + $1B which is about $17B. A recent eMarketer report estimated that Facebook would make $34B in advertising in 2017. Currently, mobile users make 54% of total users, so based on eMarketer estimates, mobile advertising should be roughly 54% X $34B or $18B. Since my estimate of $17B is very close to eMarketer's, I feel more confident that I am in the ballpark.

SYSSOC

How would you test LinkedIn's 'People You May Know' feature?

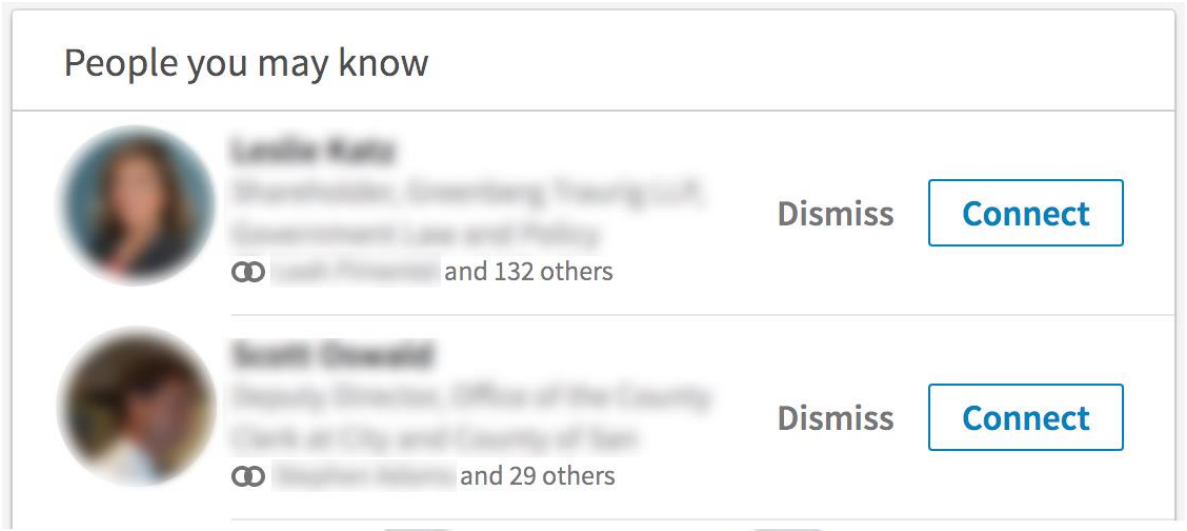
What is this question about?
This product feature testing question checks to see if you can identify which aspects of a feature to test and which techniques to use.
What is the interviewer looking for?
The interviewer is evaluating you on the following:

- Do you start with the goal?
- Do you identify what to test and why?
- Do you understand how to use common testing techniques like A/B testing?
- Is your answer structured or do you tend to talk about random points without a coherent thread?

How to structure your answer?
Structure your answer in the following way:

- Start by describing what the feature does.
- State what the goal of the feature is.
- Talk about the different components and describe what they do.
- Discuss design alternatives for those components.
- Hypothesize alternative design versions to compare with current versions. Then state how to test each hypothesis.
- Summarize your analysis and recommend which components to test.

Answer Example
INTERVIEWEE: I've seen this feature on the My Network page of LinkedIn. Before beginning, I would like to confirm my understanding of how this feature works.
INTERVIEWER: Sure.
INTERVIEWEE: The 'People You May Know' feature displays a list of connection suggestions, where the order in which the individuals appear is prioritized based on certain criteria. The order depends on how these suggestions are connected to the user. The connection could be via a mutual co-worker or co-workers, employers, schools attended, membership in organizations, or it could be an email or phone contact. And, it also seems that individuals associated with recent events that happened to the user are listed first. For example, if a user just recently added a new person to his network, than the list will display connection suggestions that are in the new person's network.
INTERVIEWER: That is a simplification but is close enough. Time overlap with a person in a company or other institution is also a key component in the prioritization. But, let's go with your description. So now, how would you test the 'People You May Know' feature?



LinkedIn's 'People You May Know' feature

INTERVIEWEE: Okay. I imagine the overall goal of this feature is to encourage LinkedIn users to grow their network by connecting to more people. Is this correct?

INTERVIEWER: Yes.

INTERVIEWEE: Ok. If I were the product manager, I would test alternative ideas against three components: the format for presenting connection suggestions, the order suggestions are listed, and the information shared about a possible connection.

Testing format of presenting connection suggestion information

Let's start with the format of presenting a connection suggestion. Currently, the feature displays a thumbnail image along with three lines of content. The first line is a name, the second line is a title, and the third line is a link that shows how this possible connection is related to the user. This link is a chain link icon, followed by the name of a user's top mutual connection and a number indicating more mutual connections. For example, the text could show *John Doe and 7 others*. Clicking on this link takes the user to a list that displays details about those mutual connections.

One question I have is if instead of using a text link to mutual connections, what if we used thumbnail images of the mutual connections? The photos would be a more visual and direct way of highlighting them. A text link such as "...and 7 others" requires an extra step to discover who those seven connections are. And, it is generally better to decrease friction by reducing the number of steps.

I would test this hypothesis using A/B testing. Version A would be the current solution with the text link, and version B would be the alternative solution using thumbnail images of the mutual connections. In case there are too many mutual connections to display, a link can be added to view additional images. If version B results in a higher number of clicks on the Connect button, then the thumbnail images version would be the better choice.

Testing Prioritization

I believe the algorithm for choosing and picking the order in which connection suggestions appear works well when the user already has many existing connections. But the algorithm may not perform as well if the user is new and does not have many connections. One question I have is whether, in the case of a newly registered user, it is better to use the user's email and phone contacts not only as connection suggestions but also as leads to additional suggested connections. Using the newly registered user's contacts may lead to more recommendations.

Again, I would use A/B testing to test this hypothesis. Version A representing the current way connection suggestions are found, and version B representing the way they are found based on email and phone contacts. This should be tested on newly registered users, not existing ones. If the Connect button in version B is clicked on with higher frequency, this would mean that showing a suggestion connection using email and phone contacts is a better approach than the default scheme for the case of newly registered users.

Currently, there is not an option to filter connection suggestions based on a mutual connection, or company, or school. One idea is to let people filter connection suggestions based on these three categories. To test this, I would run an A/B test where version A would be the current list of connection suggestions with no filter and version B would be a list of connection suggestions with the filter option. If the result of the test shows that version B received more clicks on the Connect button, then version B would be the better option.

Testing the amount of new contact information

Finally, let's talk about the information displayed about the connection suggestions. Currently, only three lines of information are shown, the name of the person, the title and a link that shows how the connection suggestion was found. The suggestion was made either because of mutual connections, mutual workplace or institution. Perhaps providing an additional data point like shared skills could tip the user's decision to click on the Connect button. So I would do an A/B test, where version A represents the current way of displaying information about a connection suggestion, and version B would have the same information as version A but with an additional line that shows shared professional skills. If the Connect button is clicked more frequently on version B, then adding shared skills would be beneficial.

To summarize, I would test the 'People You May Know' feature by first decomposing it into core elements: the format in which the connection suggestions are displayed, the prioritization criteria behind the order of connection suggestions, and the amount of information displayed about a connection suggestion. Of all the suggestions I made for testing this feature, I would prioritize testing the filter idea and including shared 'skills' to the information presented about the connection suggestion. The ability to filter data in a list has become so common that not having it may frustrate users. And, people looking for jobs are likely to want to network with others that have similar skills. Therefore, testing the appeal of these alternative suggestions would be my first priorities.

SYSSOC

How would you measure the success of Amazon Echo?

Answer Structure

The key to a product metric question is to start by defining what success means by identifying what the business goals are for the product. Once you have defined the objectives, think about what outcomes should happen in each stage of the customer journey for these objectives to be achieved. The journey towards becoming a monetizable customer usually involves these stages: awareness, acquisition, conversion, engagement, retention, and monetization. Think of measurable outcomes for each stage that would give an indication of the success or failure of the business goals.

Here is a suggested structure would be:

- Define the business goals.
- For each phase of the customer journey such as awareness, acquisition, conversion, engagement, retention and monetization, think of ways to measure customer actions that would indicate success or failure of the business goals. List each action as a metric.
- Wrap up. Summarize what the business goals and the most relevant metrics are.

Answer Example

INTERVIEWEE: I would like to start by defining what success is, which depends on what the business goals are. And then, follow with metrics to measure the achievement of those goals. The business goals should include the target buyers, which are consumer households, but also the developers of Alexa Skills, which are essential to providing a variety of service options to consumers.

My belief is that Amazon's primary goal is to grow revenue of consumer purchases from Amazon.com. While a secondary goal is to grow revenue for Amazon Web Services (AWS) by growing the number of developers building Alexa Skills.

INTERVIEWER: Yes, generally I would agree with that.

INTERVIEWEE: Ideally, the revenue from purchases through Echo should be additional revenue to that generated from selling products through the Amazon website. The way this additional revenue is likely to be generated is through:

1. **Frequent purchases that are additional to purchases through the web site**. Since the Amazon Echo is on 24×7, it is easy to see how people may find it more convenient, faster and fun to order products by voice rather than logging into the Amazon website.
2. **Increasing the number and usage of Alexa Skills**, with a view to monetize Skills' transactions via a revenue-sharing model between Amazon and developers.
3. **Increasing the number of subscriptions to Amazon Music Unlimited**. Just last year, Amazon released its Music Unlimited subscription service to compete in the music streaming market. Amazon is hoping that Echo and its family of speaker devices, along with low subscription prices, will incentivize consumers to sign up for its music service.
4. And, **increasing in the number of Amazon Prime subscribers**. Furthermore, since Amazon Prime subscribers get a slightly better price than Non-Prime subscribers, there is also an incentive for Non-Prime users to convert, which for Amazon would be a fantastic outcome.

In order to achieve these goals, consumers have to be aware of the product, buy it, use it, and continue to use it. In other words, they go through the stages of awareness, conversion/purchase, engagement and retention ending in a final monetization stage.
Let's walk through these customer journey stages to understand what kinds of interactions are involved and how to measure the success of those interactions.

Awareness

At this stage of the customer journey, the customer is learning about Amazon Echo through marketing campaigns. A good campaign should make customers curious about the product. We should see an increase across the following activities: research about the product, word-of-mouth activity, visits to the Echo website, click-through rates, blogs views, etc.
To measure the success of awareness campaigns, I would use metrics such as:

- The number of impressions per campaign across different ad types like display ads, search ads, and print ads.
- The reach of content marketing campaigns.
- An increase in the number of keyword searches after and during a campaign.
- The number of new visitors to the Echo website during and after a campaign.
- The number of repeated visitors to the Echo website during and after a campaign.
- The number of referred visitors due to word of mouth or social virality during and after a campaign.

Conversion (Purchase of Echo)

At this stage, customers are buying the Amazon Echo. Some of them will become Prime subscribers and some will not. To measure the success in the conversion of Non-Prime to Prime subscribers, it will be important at this stage to keep track of who was originally not a Prime subscriber. These are the metrics I think will be relevant at this stage:

- The number of Echo and Dot products sold per month. Is it increasing?
- The revenue from sales of Echo and Dot products per month. Is it increasing?
- How many owners are Prime and Non-Prime subscribers?

Engagement

At this stage, the customer is using Echo and we should be measuring usage activity related to our four goals: 1) whether purchases through Echo are additional to the site; 2) whether the usage of Skills is increasing; 3) whether the number of subscriptions to Amazon Music Unlimited is increasing; 4) and whether Non-Prime customers are converting to Prime.
To facilitate measuring changes in usage, purchases, and prime conversions as time goes by, we will need to segment customers into cohorts. By putting customers into cohorts based on on when they bought an Echo, we will be able to monitor changes in behavior as customers tenure increases.
Here are metrics I think we should use to measure success at this stage:
On conversion to prime subscriptions
Percentage of Non-Prime customers in a cohort that converted to Prime subscribers per month. This will indicate whether Echo has been successful in increasing Prime memberships.
On purchases
The fraction of customers in a cohort that make at least one purchase per month; this metric will indicate how frequently customers continue to use Echo to purchase products after they bought Echo.
The fraction of the same customers in a cohort that made at least one purchase every month since they bought Echo. This will indicate which fraction of the cohorts are hard-core Echo users, the more the better. Another metric I would be interested in is what their average

monthly dollar purchase is? This will indicate how habitual Echo is becoming as a channel for purchases, and if people are spending more per purchase on this new channel.

On additional purchases through Echo

What is the number of monthly purchases per household through Echo compared to monthly purchases per household through the Amazon website? Are the number of purchases through the website decreasing but increasing using Echo? This will tell us if Echo is being used as a channel for additional purchases or if Echo is replacing the Website as a channel.

Are people buying new products using Echo or buying similar products that are purchased through the website? This will tell us if Echo is incentivizing people to buy things they would otherwise not have bought through the website. Does Echo enable more impulsive buys?

On Skills

- How many Skills per month are Echo customers using?
- What are the top 10 skills per month by frequency of use?
- What are the top 10 skills per month by dollar amount?

On Amazon Music Unlimited

How many Echo buyers that were not existing music subscribers, subscribed to the Amazon Music Unlimited service after they bought an Echo? This will confirm whether Echo has been a trigger to get consumers to sign up for the new music service.

How many hours per day are Echo users listening to music? Is the average increasing or decreasing? This will indicate whether people are valuing the service.

Retention

At this stage, we are interested in knowing whether customers are likely to continue using Echo or churn. It is also important to know the interval of time between consecutive usage of Echo. The shorter the time the better for retention.

Metrics to measure retention:

- From day to day, what fraction of the same users in a cohort use Echo?
- From month to month, what fraction of the same users in a cohort use Echo at least once?
- From month to month, what fraction of the same users in a cohort use Echo to make at least one purchase?
- From month to month, what fraction of the same users in a cohort use at least one skill?
- Per month, how many days does it take on average for the a user in a cohort to reuse Echo a second time?
- Per month, how many days does it take on average for a user in a cohort to make an Amazon purchase using Echo since the beginning of the month?
- Per month, how many days does it take on average for a user in a cohort to reuse Echo to listen to music?
- How many Echoes are returned per month? Is it increasing or decreasing?

Monetization

Finally, we can quantify success in achieving the four goals: additional purchases, usage of Skills, subscriptions to Amazon Music Unlimited and Amazon Prime conversions with several metrics.

Metrics to measure monetization:

- Is the revenue from Amazon purchases through Echo decreasing or increasing?

- How much of the revenue from Amazon purchases through Echo is additional revenue or replacement revenue for purchases previously made through the Amazon website?
- If there was already a shared-revenue model in place, how much revenue was generated from Skills' transactions?
- How much revenue from Amazon Music Unlimited subscriptions is from Echo owners?
- How much revenue is from new Amazon Prime subscriptions?

And, let's not forget developers. One important objective should be to attract more developers in order to increase the number and variety of Skills offered. To measure success in this objective:

- What is the number of developers and monthly growth?
- What is the average number of Skills per developer? The more Skills developed by a single developer the more committed they are to the Echo platform.
- How much revenue was generated from hosting services and providing computing resources to Skills developers through Amazon Web Services (AWS)? An increase in revenue per developer indicates higher developer commitment.

In summary, I defined business goals with respect to selling Echo to consumers and developers. With respect to consumers, I talked about four business goals: 1) increasing additional purchases from Amazon using Echo; 2) increasing the number and usage of Skills; 3) increasing the number of subscriptions to Amazon Music Unlimited; and 4) increasing the number of Amazon Prime subscribers.

With respect to developers, I think the goals should be to increase the number and variety of Skills offered. By looking at each stage of the customer journey: awareness, conversion, engagement, retention and monetization, I defined different metrics that would help measure success towards these business goals. If I were to choose and prioritize the most important metrics to support, it would be the engagement, retention and monetization metrics, because they are direct indicators of revenue creation, which is our bottom line. In particular, I would prioritize these metrics related to increasing purchases and increasing subscriptions to Amazon Music Unlimited and Amazon Prime subscriptions, which are key revenue generating activities.

And as a final thought, as part of measuring the success of Amazon Echo, I would include device sales from competitive products like Google Home. Although device sales does not reveal insights into customer behavior, it provides a high level measure of competitiveness.

How would you increase growth and market share for Google Cloud?

What is this question about?
As a product manager, one of your responsibilities is to grow your product's revenue and market share. This product strategy interview question is testing whether you can think strategically about how to go about it.
What is the interviewer looking for?
The interviewer is evaluating you on the following:

- Are you knowledgeable about growth strategies?
- Can you think critically about what strategy could be the most fruitful given the business objective, the current product performance, the market, competition, and trends?
- Can you answer the question in a structured and organized way, or do you shoot from the hip and do so without taking a moment to think about the question?
- Do you have original ideas?
- Are you articulate or do you tend to ramble?

How to structure your answer?

- Talk about the product's strengths and weaknesses relative to competitive products.
- Who are the market leaders in this market?
- What are the market and technology drivers?
- Given the competitive state and market drivers, suggest growth strategies that touch on these points. Examples of some growth strategies are: growth by offering additional products, growth by strengthening the product's competitive position, and growth by expanding to related or new markets organically or via acquisitions.
- Summarize the opportunities for growth and your recommendations on how to exploit them.

Answer Example
INTERVIEWEE: Okay, before brainstorming about possible ways to grow revenue and market share for Google Cloud, I would like to first talk about the competitive landscape and market trends. Understanding how competition plays will help in identifying possible vectors of differentiation that Google can leverage to spur growth. And, identifying market drivers will help in identifying opportunities of adjacent or new markets to enter.
Competitors
Amazon AWS, Microsoft Azure and Google Cloud are the top players in the cloud computing market. As a first mover, Amazon has had the advantage of time; AWS market share is multiple times that of Azure and Google Cloud Platform, so that is a challenge.
Amazon has the richest and most extensive IaaS and PaaS capabilities of the three. Its strengths are in deep user management capabilities; an ecosystem of open source tools which has attracted thousands of ISV partners; a network of partners that provide application development, managed and professional services; and training and certification programs. Where AWS has weaknesses is in its complex pricing model, which is too granular, and expensive customer support.
Microsoft Azure, second in market share, has several strengths: seamless integration with enterprise on-premises infrastructure, development tools, open source technologies, competitive pricing, and a large number of existing customer relationships. But in addition to not having a complete cloud solution as AWS, Azure is not very strong in API enablement,

and lacks partners in managed services and professional services that have enough experience with Azure.

Google's strength lies in its expertise in developing and managing cloud-native applications, analytics and machine learning as well as fast virtual machine provisioning and simpler billing. Google has room for improvement in a few areas though, such as user management to provide organizations more granular and customizable access control; the need for a marketplace to license third party software and the need to increase its efforts in sales, marketing, globalization and partnerships.

Market Trends

Let's talk about what is driving the cloud computing market growth.

IoT security needs will be on the rise and cloud computing solutions for this type of security will be in demand. Last year's Dyn incident, where IoT devices were used to orchestrate a distributed denial of service (DDoS) attack on a Dyn DNS server that supported major commercial websites (Twitter, Pinterest, Reddit, Github, Spotify), shows that IoT device makers and service providers will have to increase their investment in IoT security solutions.

Data Mining and query services for vertical industries will be on the rise. Industries such as healthcare, government, finance, retail and weather forecast will look for ways to optimize customer service and operational efficiencies by adopting intelligent services that can uncover patterns, provide predictions, and answer complex questions that can only be done with advanced big data analytics technologies.

AR and VR applications will be on the rise not only for the Entertainment and Gaming market but also for the Education and Training market. These types of applications will demand vast cloud computing resources.

Advanced online collaboration tools and features will be on the rise as the number of people working in separate locations becomes the norm. This will drive the need for cloud solutions that enable deeper online collaboration. Google already has put a stake on the ground with Google Suite, and it should continue to innovate.

Ideas for Growth

Now that we have looked at the competitive landscape and market trends, I would like to brainstorm ideas for growth. I think Google Cloud should follow growth strategies based on competitiveness, acquisitions, and innovation.

Growth Based on Competitiveness

Growth could be achieved by playing on Google's competitiveness along its key differentiating technologies. One such technology is its enterprise APIs. The acquisition of Apigee was a step in the right direction; it put Google ahead of AWS and MSFT in the API software integration space. Google should continue to expand features on these platform or acquire additional startups in this space.

Google could also use its differentiated technologies like Machine Learning, AI, and VR/AR to expand into related markets. For example, it could expand into the AR/VR Training and Education software market by leveraging its strong relationships with schools and universities.

Google is one of the few large technology companies that has the computing power to provide data mining and query services that vertical industries with huge Big Data needs can benefit from. In healthcare, Google is already partnering with schools such as Stanford Medicine to provide a genomics service mostly for research, but it should try to expand the service for commercial purposes. Microsoft just announced its Healthcare Next initiative to bring AI to doctors and hospitals, and Google should do the same.

Partnerships are one of the weak points of Google Cloud. Partnerships with application developers, managed services and professional services are key to growing the number of customers. More investment in these efforts is needed. Furthermore, to be able to increase the number of enterprise customers, it needs to have an application marketplace, just like AWS and Azure has.

And given that AWS and Azure are complex platforms to develop on and manage, Google could strive to provide ease of use and training programs to evangelize its platform to customers and developers.

Growth Based on Acquisitions

As IoT security becomes a critical issue for all enterprises, Google should start working towards becoming the leader in this space. Acquisitions in this space would make it faster to acquire the technology and skills in IoT security, which would also help promote Google IoT related businesses like Nest, Android Wearables and Google's driverless technology.

Growth Based on Innovation

Advanced online collaboration is an interesting market too. Google dominates in the Education space with Google Suite, and the SMB space, and needs to grow its enterprise presence. The online collaboration market, however, is a more mature market and growing slower than other markets. I think growing this market through product innovation, customer relationships and partnerships, as opposed to acquisitions, would result in a better bang for your buck. Acquisitions are expensive and I don't think this market is growing fast enough to offer a quick return.

In summary, I think Google Cloud can grow revenue and market share through strategies that include competitive differentiation, acquisitions, partnerships and innovation. In particular, Google could leverage its strengths in Big data analytics, machine learning and AR/VR, to expand into related markets like Education & Training, industry verticals such as Healthcare, Finance and Government. Acquiring selective startups in IoT security would strengthen Google's leadership in this market, which is likely to grow rapidly. And, it should continue to innovate in the online collaboration space. Of course Google should continue to acquire technologies to catch up with Amazon's offerings, but faster growth will come if Google leverages its differentiating technologies and skills to become number one in areas of cloud computing that its competitors can't easily copy.

SYSSOC

How would you implement a feature that uses location in Twitter?

Question

You are a product manager for Twitter and you are asked to design a feature that uses the location of tweets. What would you propose?

Answer Structure

This is a typical product design question. One useful structure to apply to your thinking when tackling this type of question is the CIRCLES Method™ from Lewis Lin's book *Decode and Conquer*, which boils down to:

- Ask clarifying questions to narrow the scope and to understand what the goal is
- Who is the target user(s)
- What are the use cases
- Prioritize use cases
- Brainstorm solutions
- Evaluate the tradeoffs of solutions
- Summarize: which solution you would recommend; recap what the solution does, why is it beneficial and why you prefer this solution

This is just a guideline, use it judiciously.

Answer Example

INTERVIEWEE: What is the goal of the feature?

INTERVIEWER: The goal is to increase the frequency of use and engagement of Twitter users. We think this will increase the attractiveness of the platform to marketers/advertisers.

INTERVIEWEE: There is a Connect feature in the Twitter app that lists local events under the label Popular Near You. It lists, for example local San Francisco news/events and and public city services like BART, the San Francisco Police or publications like 7×7. So, is this feature not working the way you expected?

INTERVIEWER: Well, we think we can do better, and would like to get more ideas from you.

INTERVIEWEE: Ok. Would you like me to provide ideas for desktop and mobile?

INTERVIEWER: Let's narrow it down to the mobile app.

INTERVIEWEE: Ok. Could I take a few minutes to think about possible solutions?

INTERVIEWER: Please do.

The interviewee starts by thinking who the possible Users are, which will help him think about use cases and answer the rest of the questions.

Twitter Users:

- News readers (national/international news)
- Celebrity followers
- Technology followers
- Influencers/Thought leader followers
- Entertainment (Movies, Hollywood gossip) news followers

INTERVIEWEE: I would like to start off by talking about the different types of Twitter users and why they use Twitter. This will help me identify who the most valuable users are and prioritize their needs. Then, I will brainstorm how location can be used to enhance the user's experience, provide some solutions, and wrap up with recommendations.

INTERVIEWER: Sounds good.

INTERVIEWEE:

User types

In general, Twitter users use the platform to read news that is of particular interest to them. For example:

- News junkies
- Celebrity followers
- Technology news followers
- Followers of influencers or thought leaders
- Followers of entertainment news like latest Movies, Hollywood gossip
- Friends followers

I think the heaviest users from these groups are:

1. **Followers of entertainment news**—I think most people go to Twitter to be entertained, such as reading about the latest hollywood news or celebrity gossip.
2. **News junkies**—people are getting their news from social networks like Twitter more and more rather than newspapers.
3. **Celebrity followers**—many millennials follow celebrities.
4. **Technology news junkies**—mostly happening in states with large concentration of technology companies.
5. **Influencer followers**—people following thought leaders in business and technology mostly.
6. **Friends followers**

I would prioritize the needs of the top three users, since they are the heaviest users.

Use Cases

Now, I would like to analyze how these users use Twitter, then think how location can be used to enhance their experience.

- **Entertainment followers**—these users are probably doing searches about movies or tv shows, actors or seasonal events like the Oscars.
- **News junkies**—these users already follow specific news sources like the New York Times, BBC for international news or the San Francisco Chronicle for local news for example; and they are also doing searches for tweets about recent international/national events people are talking about.
- **Celebrity followers**—these users are reading what their favorite celebrities are tweeting about.

Brainstorm Solutions

Now, I am going to brainstorm some possible new features that use location to increase engagement and frequency of use of these users.

Three ideas are:

1. Extend the **Popular Near You** feature to include filters that allow users to filter news/events by a topic of interest.
2. Create a new **Local tab** that lets users find tweets about local news or events that can be filtered by a topic of interest.
3. Add a **Marketplace tab** similar to what Facebook has that enables users to find things or services for sale near them.

Of the three ideas, I would pick the one that is more in line with the goals of the users, engages them more and is easier to implement. Extending the Popular Near You feature and creating a new Local tab feature are more aligned with the goals of the top Twitter users, which is to be entertained and read about news and interesting events. The Marketplace tab would only interest people that are in the market to buy or sell something local, which I think is less aligned with the goals of these users. Between the Popular Near You feature and the

new Local tab feature I am proposing, I think the new Local tab would be the best option. A Local tab would be used more frequently because it would be visible on the app tab bar as opposed to buried under the Connect feature as the Popular Near You label. And, it would be easy to implement because it is similar to a regular feed except filtered by location and topics.

INTERVIEWER: Okay, sounds good. So how would you implement it?

INTERVIEWEE: Okay, before talking about implementation, I would like to start by describing how I think location information can be used to enhance the top users' experience.

- For **Entertainment followers**, location can be used to enable them to find local events happening that day with the option of filtering by different types of events, such as movies, concerts, bar events, and coffee house events. It could also enable them to buy event tickets.
- For **News followers**, location can be used to enable them to filter local news, or automatically show tweets about local news related to tweets the user has retweeted or liked.
- For **Celebrity followers**, location could be used to enable them to find which celebrities are performing in town that day and enable them to buy event tickets.

Let's talk about implementation now. The Local tab would present a new screen after the user taps it, and like with the Moments tab, the categories of Entertainment, News, and Celebrity can be presented on a horizontal bar at the top of the screen. The user can select any of these three topics to display related tweets that are happening within the city the user lives. In the case of Entertainment, when the user selects this topic, the sub-filters such as movies, theater, coffee bars, and concerts, would replace the contents of the horizontal bar. A back button would need to be displayed, perhaps as the first button on the list of sub-filters to enable the user to return to the main topics.

Since there would be space for additional main topics on the top horizontal bar, the Technology News, Influencers and Friends topics could be included as additional main topics.

However, one drawback I see with some categories like Celebrities, is that there may not be too many tweets to show at the local level, since Celebrities tend to hang out only in very cosmopolitan/international cities. A way to deal with this problem is to display this topic only in cosmopolitan cities like New York, Los Angeles, Chicago, and others.

In terms of monetization, I think that the Entertainment events topic would attract many of the local businesses to promote or advertise within this Local feature. People tapping on the Local tab would already be interested in events in their local sphere and more likely to buy tickets for local events. Twitter could charge these businesses a percentage of the ticket prices bought through its Local feature to generate revenue.

In summary, I brainstormed three possible solutions that use tweet locations to enhance the experience of top heavy users of Twitter: an extension to the current Popular Near You feature, a Local tab, and a Marketplace tab. I recommend the implementation of the Local tab because its function would be more aligned with the main goals of the most frequent users and would perform better than the current Popular Near You feature, which is not visible. An additional upside of the Local tab feature, besides increasing frequency of use and engagement, is that it can be monetized through a profit margin sharing arrangement with event organizers when users buy tickets.